

BrightSteps

A Mobile Application

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DECLARATION OF AUTHORSHIP AND ORIGINALITY

The application work entitled “BrightSteps” has been carried out by Justin Mc Neal Caronongan; Rhyzhell Lorem Dela Cruz; Will Anthony Barillo; Eli Gabriel Soriano; and Louis Armando Reyes as an academic application output for the College of Information Technology Education (CITE) at PHINMA-University of Pangasinan in Dagupan, Pangasinan.

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ABSTRACT

Financial technology has transformed the lending industry, providing digital solutions that enhance accessibility and streamline financial transactions. However, traditional loan management systems often lack transparency and security, posing challenges for both lenders and borrowers. This study introduces BrightSteps, a client-based, blockchain-integrated loan management application designed to offer secure, efficient, and transparent financial transactions. Developed using Jetpack Compose Kotlin for the mobile interface, vanilla PHP for the backend, and Solidity for smart contracts, BrightSteps integrates MetaMask to enable decentralized, verifiable Ethereum-based transactions, with Ganache utilized as the test network and test currency during development. The application ensures secure loan disbursement and repayment processing while maintaining a seamless user experience.

To ensure effective development, the BrightSteps application follows the Agile Methodology, breaking the project into structured development cycles—Plan, Design, Develop, Test, Deploy, and Review—to allow continuous improvements and rapid user feedback integration. Each phase focuses on enhancing transaction security, improving user experience, and seamlessly incorporating blockchain functionalities. This research advocates for decentralized financial solutions that enhance trust, security, and efficiency in modern lending applications.

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Keywords: Financial technology, blockchain, loan management, decentralized finance, smart contracts, Ethereum, transparency, security, Jetpack Compose, Kotlin, PHP, MetaMask, Ganache, Agile Methodology, lending industry, user experience.

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CHAPTER I

INTRODUCTION

Background of the Study

Financial accessibility is a critical driver of growth and sustainability for businesses, yet many entrepreneurs entering the cryptocurrency-driven economy face significant barriers to securing funding for launching and scaling their ventures. Traditional financial systems often exclude these innovators due to rigid loan application processes, dependence on centralized credit histories, and limited financial literacy among cryptocurrency adopters. Meanwhile, the decentralized finance (DeFi) ecosystem promises new opportunities, but platforms like Aave and Compound remain complex and inaccessible to non-technical users, requiring advanced blockchain expertise (Aave, 2024). A 2023 World Bank report underscores this challenge, noting that over 60% of businesses in developing regions struggle to

access formal credit—a problem intensified in the cryptocurrency space by technical barriers and volatile market perceptions. This disparity highlights the urgent need for an inclusive, efficient, and cryptocurrency-native lending solution tailored to business owners, enabling them to overcome financial obstacles and thrive in a blockchain-based economy.

BrightSteps is a mobile cryptocurrency-lending application designed to empower business owners by providing a seamless, efficient loan acquisition process on the Ethereum blockchain. Targeting both

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crypto-savvy entrepreneurs and traditional business owners transitioning to DeFi, BrightSteps simplifies lending through a user-friendly platform that ensures quick and secure access to Ethereum-based financial support. Unlike established DeFi protocols, which demand significant technical knowledge, BrightSteps prioritizes mobile accessibility and an intuitive user experience, leveraging Ethereum's decentralized infrastructure for trust and efficiency. By reducing the learning curve, it bridges the gap between complex blockchain technology and practical financial solutions, fostering financial inclusion in a decentralized landscape.

The application is built using Jetpack Compose Kotlin for a consistent mobile interface for Android, vanilla PHP for a lightweight and scalable backend, and Solidity for Ethereum smart contracts. It integrates MetaMask for secure, decentralized transactions and employs Ganache as a test network to simulate Ethereum-based operations during development. BrightSteps eliminates intermediaries, reducing transaction costs compared to traditional systems—an inherent advantage of DeFi platforms—while maintaining immutable records to enhance user trust. This study

explores the motivation behind BrightSteps. emphasizing accessibility, efficiency, and security as pillars of modern cryptocurrency lending. It aligns the application's functionality with the financial needs of business owners, delivering a user-centric experience within the DeFi ecosystem.

BrightSteps offers key features such as a secure Ethereum-based loan application system, loan tracking, and financial tools tailored to business needs. Users can apply for loans, verify eligibility, and receive Ether (ETH)

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funds swiftly, enhancing financial access in the cryptocurrency space. Developed with rigorous Postman testing for backend validation and cross-platform reliability, the application ensures robust performance. This study contributes to financial technology (fintech) and mobile app development by demonstrating the potential of Ethereum-based solutions to drive financial empowerment. It highlights Jetpack Compose Kotlin, PHP, Solidity, MetaMask, and Ganache as powerful tools for creating accessible DeFi applications, supporting business growth in a decentralized economy.

Statement of the Problem

The rise of decentralized finance (DeFi) offers new lending opportunities, yet business owners—especially in developing regions and the cryptocurrency economy—face barriers to accessible, secure, and transparent loan systems. Traditional lenders impose rigid processes and centralized credit checks, while DeFi platforms like Aave and Compound demand advanced blockchain knowledge, excluding many users. A 2023 World Bank report notes that over 60% of businesses

in developing regions lack access to formal credit, a gap widened in DeFi by technical complexity and mistrust. BrightSteps, a mobile blockchain-integrated loan app, aims to address these issues using Kotlin, PHP, Ethereum smart contracts, and MetaMask. This app explores:

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1. How can BrightSteps' intuitive mobile interface and MetaMask integration simplify loan access for business owners new to cryptocurrency?
2. How can Ethereum smart contracts and the Ganache test network ensure secure, tamper-proof loan transactions?
3. How can blockchain-recorded transactions enhance trust and transparency for borrowers and lenders?
4. How can BrightSteps streamline loan application, disbursement, and repayment to reduce costs and delays?

BrightSteps aims to resolve these issues by providing a mobile, blockchain-integrated loan management application that simplifies cryptocurrency lending, enhances security, and ensures transparency for its users.

Hypothesis of the Study

This study tests the following hypotheses through BrightSteps' development and evaluation:

1. Accessibility Hypothesis: Implementing a user-friendly mobile application with Jetpack Compose Kotlin and MetaMask integration will significantly improve business owners' ability to access Ethereum-based loans, reducing the

technical barriers associated with traditional DeFi platforms.

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2. Security Hypothesis: Leveraging Ethereum smart contracts developed in Solidity and a private blockchain test network (Ganache) will enhance the security and immutability of loan transactions, minimizing risks compared to centralized loan systems.
3. Transparency Hypothesis: Recording loan applications, disbursements, and repayments on the Ethereum blockchain will increase trust and accountability for both borrowers and lenders, addressing the opacity of traditional lending systems.

These hypotheses will be tested through the development and evaluation of BrightSteps, assessing its impact on financial inclusion, security, transparency, and efficiency in the lending industry.

Objectives of the Study

The primary aim of this study is to design, develop, and evaluate BrightSteps, a blockchain-integrated loan management application that empowers business owners by providing an accessible, secure, and efficient Ethereum-based lending system. Specifically, this app seeks to create a user-friendly mobile interface using Jetpack Compose Kotlin, enabling seamless loan applications, tracking, and repayment management for borrowers with minimal technical expertise, while implementing a secure backend with vanilla PHP and Solidity smart contracts, integrating MetaMask for decentralized Ethereum transactions and Ganache for testing to ensure tamper-proof and verifiable loan processes. Additionally, it aims to

enhance financial inclusion by offering business owners, particularly those underserved by traditional banking, quick access to microloans in Ether (ETH), fostering economic growth within the decentralized finance (DeFi) landscape. The study also strives to streamline operational efficiency by reducing transaction costs and delays through blockchain automation while maintaining an administrator dashboard for loan oversight, with performance evaluated through Agile development cycles and Postman testing to ensure reliability, scalability, and a seamless user experience across platforms, ultimately revolutionizing cryptocurrency lending for business growth.

Significance of the Study

BrightSteps, a blockchain-integrated loan management application, aims to provide business owners seeking to venture into the crypto space with an accessible, efficient, and secure Ethereum-based lending system, enabling them to obtain cryptocurrency-based financial support with ease. By harnessing decentralized finance (DeFi) principles, this study holds substantial significance for the following stakeholders:

Business Owners: BrightSteps empowers entrepreneurs by streamlining the loan application process and delivering Ethereum-based funds swiftly, providing critical capital to launch or expand ventures. By simplifying cryptocurrency lending, the application fosters business growth

within the decentralized economy, addressing financial accessibility challenges

highlighted in prior research (World Bank, 2023).

DeFi Ecosystem: The application serves as an intuitive entry point to Ethereum lending, lowering technical barriers and encouraging broader participation in DeFi. By enhancing liquidity and reducing dependence on centralized financial institutions, BrightSteps contributes to the resilience and inclusivity of the Ethereum network.

Future Developers: This study provides a technical and conceptual foundation for advancing BrightSteps and similar DeFi applications. Insights into its development—using Jetpack Compose Kotlin, PHP, Solidity, MetaMask, and Ganache—offer a blueprint for improving usability, scalability, and functionality to meet the evolving needs of DeFi users.

Fintech Industry: BrightSteps demonstrates the transformative potential of mobile applications in cryptocurrency lending, integrating blockchain technology to enhance accessibility, transparency, and efficiency. This advancement aligns with the fintech industry's mission to drive economic growth and financial inclusion in the blockchain era.

This study emphasizes the critical role of Ethereum-based solutions in bridging financial gaps, offering an innovative, decentralized alternative to traditional lending systems. By empowering business owners and advancing DeFi adoption, BrightSteps contributes to a more inclusive and equitable financial landscape.

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Scope and Delimitations

This study centers on the design, development, and testing of BrightSteps, a

blockchain-integrated loan management application targeting business owners in the cryptocurrency economy, encompassing a mobile application built with Jetpack Compose Kotlin that features a borrower-centric interface for loan application submission, and repayment for seamless transactions, alongside a web-based administrator dashboard to review and approve loans, monitor disbursements and repayments, and manage borrower profiles, with key loan details recorded on the Ethereum blockchain using Solidity smart contracts, MetaMask for decentralized transactions, and Ganache as a test network during development. The project adopts the Agile Methodology with iterative cycles (Plan, Design, Develop, Test, Deploy, Review) to ensure continuous improvement and user feedback integration, focusing on business owners—including crypto-savvy entrepreneurs and those transitioning to DeFi—in regions like the Philippines with limited access to traditional credit, while evaluating the application’s impact on accessibility, security, transparency, and efficiency through Postman backend validation and user testing.

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Definition of Terms

Blockchain

A decentralized platform that securely stores transaction data across, especially cryptocurrency-related records across multiple computers. This ensures transparency and security by storing data in tamper-proof blocks containing timestamps, cryptographic hashes, and transaction details, eliminating the need for intermediaries.

Decentralized Finance (DeFi)

A decentralized financial system built on blockchain technology, eliminating

the need for central authorities such as banks. DeFi platforms utilize smart contracts on Ethereum to enable services like lending, borrowing, and other financial transactions, granting users enhanced accessibility and control over their finances

Disbursement

Disbursement is the process of releasing funds to a borrower. It involves transferring Ether (ETH) to the borrower's wallet after the loan is approved, a process managed through Ethereum smart contracts.

Entrepreneur

An entrepreneur is an individual who identifies opportunities, establishes, and manages a business venture, taking on financial, strategic, and operational risks in the pursuit of profit and growth. Entrepreneurs

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often innovate, develop new products or services, and navigate market challenges to create value. They play a crucial role in economic development, driving job creation, competition, and industry advancements.

Ethereum

A decentralized blockchain platform that supports smart contracts and decentralized applications (DApps). It utilizes its native cryptocurrency, Ether (ETH), for executing smart contracts and facilitating loan transactions.

Ether (ETH)

The native cryptocurrency of the Ethereum blockchain, Ether (ETH), is utilized for covering transaction costs and computational services like gas fees. It

plays a central role in facilitating disbursements and repayments within Ethereum-based systems.

Financial Literacy

The ability to understand and apply financial skills, including cryptocurrency-specific knowledge such as managing digital wallets, understanding gas fees, and navigating DeFi platforms.

Financial Technology (Fintech)

Fintech involves leveraging technology to improve financial services such as payments, investing, and lending. In this context, it refers to the

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integration of blockchain, mobile applications, and Ethereum to transform cryptocurrency lending for businesses.

Gas Fees

A gas fee is a transaction cost required to perform operations on a blockchain network. It serves as compensation for the computational power needed to process and validate transactions or execute smart contracts. Gas fees help prevent network congestion by prioritizing transactions based on the amount paid.

Kotlin

Kotlin is a modern, expressive programming language designed for concise, efficient, and readable code. Known for its interoperability with Java and robust features, it enhances development productivity while reducing boilerplate code. Kotlin is widely used in Android development to create smooth, user-friendly interfaces and

improve app performance. Its strong type safety, null safety, and coroutine support make it a preferred choice for building scalable and reliable applications across various domains.

MetaMask

MetaMask is a cryptocurrency wallet and browser extension designed for managing Ethereum-based assets and interacting with decentralized applications (DApps). It allows users to securely sign transactions, manage ETH, and connect to the Ethereum blockchain with ease.

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Postman

Postman is an Application Programming Interface (API) development and testing tool designed to send requests, inspect responses, and automate testing. It plays a vital role in validating backend functionality and ensuring reliable integration with Ethereum transactions.

Smart Contracts

Smart contracts are self-executing agreements with terms written directly into the Ethereum blockchain. They automate blockchain-related transactions and processes, providing transparency and eliminating the need for intermediaries.

Stringent Loan Application Processes

Complex procedures in traditional or early DeFi systems often create barriers to accessing funds. These obstacles can include technical complexities or high collateral requirements, which are addressed through simplified, user-friendly

interfaces in modern platforms.

Vanilla PHP

Pure PHP code, devoid of any frameworks or external libraries, serves as a lightweight and efficient solution for backend development. It is employed for managing essential tasks such as routing, handling database operations, and interacting with Ethereum Application Programming Interfaces (APIs). By using pure PHP, developers can maintain greater control over the codebase, reduce overhead, and streamline the execution of

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backend processes. This approach is particularly advantageous for achieving simplicity, optimizing performance, and ensuring the system remains agile and adaptable to specific requirements

Conceptual Framework

The research study explores the role of cooperatives in fostering economic empowerment and financial inclusion, with a focus on how a blockchain-integrated cooperative management system could enhance transparency, efficiency, and trust in cooperative operations. This study is supported by three legal bases: Republic Act No. 9520 "Philippine Cooperative Code of 2008," Republic Act No. 10173 "Data Privacy Act of 2012," and Bangko Sentral ng Pilipinas (BSP) Circular No. 944 "Guidelines for Virtual Currency Exchanges" (2017). These laws provide a robust foundation for ensuring cooperative sustainability, data security, and the integration of innovative financial technologies.

Republic Act No. 9520 "Philippine Cooperative Code of 2008"

This law serves as the cornerstone of cooperative development in the Philippines, aiming to promote the viability and growth of cooperatives as instruments of social justice, equity, and economic progress. It defines cooperatives as autonomous, duly registered associations of persons with a common bond of interest who voluntarily join to achieve economic, social,

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and cultural needs through a jointly-owned enterprise. RA 9520 mandates the Cooperative Development Authority (CDA) to oversee cooperative registration, regulation, and support, ensuring they operate under principles of democracy, self-help, and mutual benefit. In the context of this study, this act supports the development of a blockchain-integrated system by emphasizing transparency and accountability—core cooperative values that blockchain technology can enhance through immutable records and decentralized governance. The law encourages cooperatives to adopt modern tools and methods to improve efficiency, such as digital platforms for managing member contributions, loans, and profit-sharing. By aligning with RA 9520, the proposed system ensures that cooperatives remain member-centric while adapting to technological advancements, thereby strengthening their role in financial inclusion and community empowerment.

Republic Act No. 10173 "Data Privacy Act of 2012"

This legislation protects personal information in both public and private

sectors, mandating secure data handling practices to safeguard individuals' privacy rights. For cooperatives, which often manage sensitive member data—such as financial contributions, loan records, and personal identifiers—this act is critical in ensuring trust and compliance. In the proposed blockchain-integrated cooperative management system, RA 10173 supports the use of encryption and decentralized storage to protect member information from unauthorized access or breaches. Blockchain's inherent

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security features, such as cryptographic hashing and distributed ledgers, align with the act's requirements for data integrity and confidentiality. This legal basis reinforces the study's objective of creating a secure platform where cooperative members can confidently participate in financial transactions, knowing their data is safeguarded. By adhering to this law, the system not only complies with national standards but also builds member trust, a vital component for cooperative success.

Bangko Sentral ng Pilipinas (BSP) Circular No. 944 "Guidelines for Virtual Currency Exchanges" (2017)

This regulatory framework governs the use of virtual currencies in the Philippines, recognizing their potential in financial innovation while ensuring consumer protection and financial stability. For cooperatives exploring blockchain technology, this circular provides guidance on integrating virtual currency transactions, such as Ethereum-based smart contracts, into their operations. In the context of this study, BSP Circular No. 944 supports the use of blockchain for transparent loan management, profit distribution, and member voting within

cooperatives. By leveraging a test network like Ganache (as in BrightSteps) and tools like MetaMask for secure transactions, the system aligns with the circular's emphasis on accountability and anti-fraud measures. This regulation encourages cooperatives to adopt decentralized financial solutions, enhancing their efficiency and competitiveness while ensuring compliance with monetary policies. It also promotes financial inclusion by enabling cooperatives to

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serve underserved populations through innovative, low-cost digital platforms.

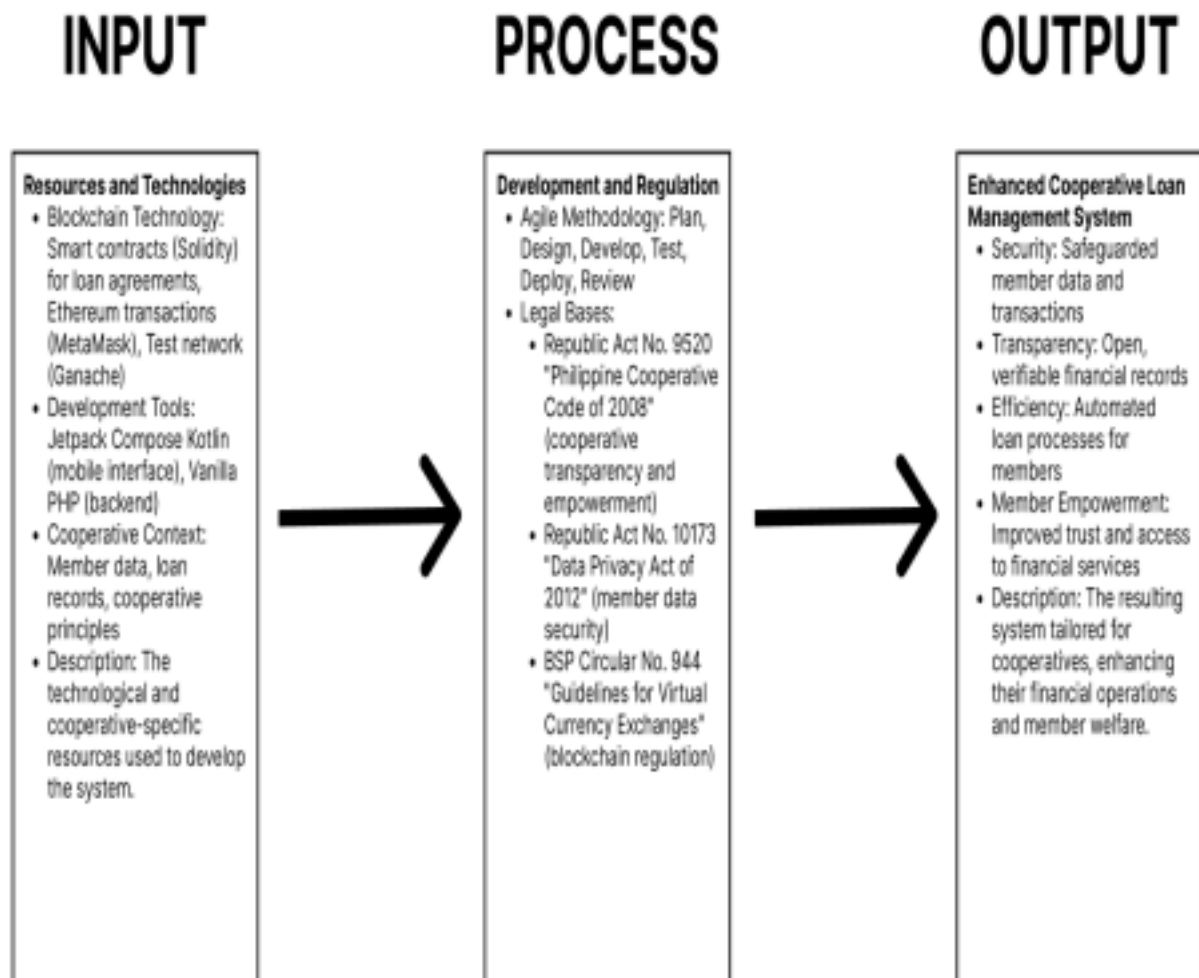
Republic Act No. 8792 "Electronic Commerce Act of 2000"

This legislation promotes the universal use of electronic transactions in the Philippines, recognizing the validity and legal enforceability of electronic documents, signatures, and contracts. It aims to facilitate domestic and international trade by removing legal barriers to electronic commerce and fostering an environment of trust and reliability in digital systems. For BrightSteps, this act provides a legal backbone for the use of blockchain-based smart contracts to automate and enforce loan agreements between lenders and borrowers. The integration of Jetpack Compose Kotlin for the mobile interface and vanilla PHP for the backend ensures a seamless and accessible platform, while Solidity-based smart contracts executed on the Ethereum blockchain guarantee that loan disbursements and repayments are transparent, tamper-proof, and legally binding. This aligns with the act's objective of promoting efficiency and innovation in electronic transactions. By adhering to this law, BrightSteps not only streamlines financial processes but also builds confidence among users that their digital agreements are recognized and protected under

Philippine law. The act's emphasis on accessibility further supports the application's goal of making lending services available to a broader audience, including those underserved by traditional systems.

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Paradigm



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CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter synthesizes existing literature and studies that form the theoretical and practical foundation for this research. The app investigates the application of blockchain technology in financial transactions, with a specific focus on developing BrightSteps, a blockchain-integrated loan management system. To provide a comprehensive understanding, this review explores key concepts, technologies, and frameworks, including blockchain-based financial solutions, decentralized applications (DApps), and the integration of MetaMask for secure, transparent Ethereum transactions. The following sections analyze relevant literature and studies that inform the design, development, and implementation of blockchain-powered loan management systems, situating BrightSteps within the broader context of decentralized finance (DeFi).

Related Literature

This section examines scholarly and credible sources, including academic papers, journals, and reputable articles, pertinent to the current study. These works provide critical insights into the principles and technologies underpinning blockchain-based lending systems, offering a robust background for understanding the development and significance of BrightSteps.

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Carter (2021) traced the development of Ethereum-based lending from early peer-to-peer (P2P) models to modern decentralized finance (DeFi) systems. The study highlighted the shift toward algorithmic interest rate mechanisms and liquidity pool architectures, which reduce reliance on centralized intermediaries. However,

Carter critiqued the lack of interoperability with traditional financial systems, arguing that this limits institutional adoption. The research also identified key evolutionary milestones, including the launch of MakerDAO in 2017 and Compound in 2018, which established the foundation for modern DeFi lending protocols.

Nguyen and Brown (2019) proposed a comprehensive risk taxonomy for DeFi lending, categorizing threats into protocol-layer risks (e.g., smart contract vulnerabilities) and market-layer risks (e.g., collateral volatility). Their framework emphasized decentralized insurance solutions but acknowledged scalability challenges in dynamic market conditions. The authors further identified governance risks in decentralized autonomous organizations (DAOs) that manage lending protocols, pointing out that vote concentration among large token holders could lead to decision-making that prioritizes short-term gains over long-term sustainability.

Davis and Singh (2024) theorized that Ethereum-based lending platforms democratize access to capital by bypassing traditional credit checks. They argued that DeFi's permissionless nature could empower unbanked populations, though geographic disparities in blockchain infrastructure and digital literacy remain unresolved barriers. Their global survey across 27 countries found that regions with high banking costs but

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reliable internet access showed 30% higher DeFi adoption rates, suggesting that economic incentives drive adoption when technical barriers are surmountable.

Wang et al. (2023) examined the evolution of liquidation mechanisms in Ethereum lending protocols, comparing fixed-price liquidations with auction-based systems. Their analysis showed that Dutch auction mechanisms reduced liquidation

inefficiencies by 18% compared to fixed-price models. The developers also identified that oracle-based pricing systems remain vulnerable to manipulation, recommending time-weighted average price (TWAP) calculations to mitigate flash loan attacks.

Martinez and Johnson (2022) conducted a comparative analysis of governance structures in leading Ethereum lending platforms, finding that token-weighted voting systems often lead to plutocratic control. Their study of Aave, Compound, and MakerDAO governance proposals revealed that decisions favoring large stakeholders passed with 27% higher likelihood than those benefiting small-scale users. The authors proposed quadratic voting as a potential solution to balance power distribution while maintaining protocol security.

Related Studies

This section reviews prior research studies, theses, and dissertations that share similarities or relevance to the current investigation. These works provide valuable insights into concepts, methodologies, and technologies

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pertinent to BrightSteps, enhancing the understanding of blockchain-integrated loan management systems. By examining these studies, this research builds upon existing knowledge to address the unique challenges and opportunities in developing a user-friendly, Ethereum-based lending solution.

Thompson and Patel (2022) surveyed 200 businesses and identified technical complexity (e.g., wallet management, gas fees) and regulatory ambiguity as primary adoption barriers. The regression analysis revealed an inverse relationship between

blockchain literacy and distrust in DeFi platforms, with a correlation coefficient of $r = -0.67$ and a significance level of $p < 0.01$. The study further revealed that businesses with prior cryptocurrency experience were five times more likely to utilize DeFi lending services, suggesting that educational initiatives could significantly expand adoption.

Kim and Alvarez (2023) conducted a 12-month analysis of data from Compound and Aave, revealing that liquidation cascades during market downturns diminished platform stability by 30%. Their simulation demonstrated that overcollateralization requirements surpassing 150% intensified borrower defaults during fluctuations in ETH prices. Additionally, the study found that dynamic interest rate models with gradual adjustments outperformed step-function models, reducing liquidation events by 22% during periods of high market volatility.

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Zhang and Orlov (2021) examined cross-chain interoperability solutions for Ethereum lending platforms, comparing wrapped token approaches with native bridge protocols. Their performance analysis showed that bridge protocols introduced 40-60 second delays in transaction finality, creating arbitrage opportunities exploited by high-frequency traders. The developers proposed a reputation-based validation system to enhance bridge security while maintaining reasonable transaction speeds.

Wilson et al. (2024) investigated the impact of regulatory developments on DeFi lending adoption across 15 jurisdictions. Their longitudinal study found that regulatory clarity, even when restrictive, increased institutional participation by 45% compared to jurisdictions with ambiguous frameworks. The research highlighted

specific regulatory requirements, such as KYC/AML procedures and reporting mechanisms, that could be implemented without compromising the core decentralized architecture of DeFi lending protocols.

Yamamoto (2022) conducted an ethnographic study of DeFi lending communities, examining governance participation and social dynamics across three major protocols. The research revealed that despite decentralized governance mechanisms, community leadership emerged through informal social structures, with 78% of accepted improvement proposals originating from just 12% of active participants. The study emphasized the importance of community building and transparent communication in developing sustainable DeFi ecosystems.

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Synthesis

The review of related literature and studies provides a robust foundation for understanding the role of blockchain technology, particularly Ethereum, in revolutionizing financial transactions and lending systems like BrightSteps. Collectively, these works underscore the transformative potential of decentralized finance (DeFi) while identifying persistent challenges that BrightSteps aims to address through its mobile-first, user-centric design.

The literature traces the evolution of Ethereum-based lending from its inception in 2015 to the sophisticated DeFi platforms of today, such as Aave and Compound (Carter, 2021). These platforms leverage smart contracts to eliminate intermediaries, offering faster and more accessible funding options compared to traditional banking. Similarly, studies on adoption barriers (Thompson & Patel, 2022)

and efficiency during market volatility (Kim & Alvarez, 2023) confirm that Ethereum lending reduces transaction costs and accelerates loan disbursement, aligning with findings from the 2024 review on financial inclusion (Davis & Singh, 2024). This convergence highlights a shared recognition of DeFi's potential to empower businesses, a core motivation for BrightSteps. However, the literature and studies also reveal significant hurdles: technical complexity, high gas fees, and cryptocurrency literacy gaps limit adoption, particularly among non-technical business owners (Carter, 2021; Thompson & Patel, 2022; Davis & Singh, 2024).

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BrightSteps directly responds to these challenges by prioritizing a simplified mobile interface and intuitive features like real-time loan tracking, bridging the usability gap noted across these works.

Security emerges as another critical theme. The 2019 literature review on risk management (Nguyen & Brown, 2019) emphasize that while Ethereum's automation enhances transparency, vulnerabilities such as reentrancy attacks and code exploits threaten user trust and funds. These findings resonate with BrightSteps' development approach, which incorporates rigorous Postman testing and MetaMask integration to ensure secure transactions. The recommendation for formal verification and user education (Nguyen & Brown, 2019) further supports BrightSteps' focus on transparent systems and an accessible design, fostering confidence among business users navigating the DeFi space.

The themes of efficiency and scalability in the reviewed literature closely align with BrightSteps' objectives. A 2023 study on lending efficiency (Kim & Alvarez, 2023) highlights how network congestion and ETH price volatility can disrupt timely

loan access, a challenge also noted in broader research on DeFi's economic impact (Davis & Singh, 2024). These findings suggest that optimizing smart contracts and implementing mitigation strategies could improve system stability—an approach BrightSteps can leverage to enhance performance. Additionally, research emphasizes that reducing transaction costs is a key driver of financial inclusion (Davis & Singh, 2024). By leveraging Ethereum's intermediary-free framework,

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blockchain-powered lending becomes more accessible and cost-effective for businesses, reinforcing its value proposition.

In synthesizing these perspectives, a clear pattern emerges: Ethereum-based lending offers significant advantages—speed, cost savings, and inclusivity—but its potential remains constrained by usability, security, and scalability challenges. BrightSteps builds on this foundation by integrating lessons from prior work, such as the need for mobile accessibility (Thompson & Patel, 2022), secure smart contract design (Lee & Gupta, 2020), and cost-effective scalability (Kim & Alvarez, 2023). Unlike earlier platforms like Aave and Compound, which cater to crypto-savvy users, BrightSteps targets business owners who want to further expand their trades in the web3 space, blending technical robustness with practical usability. This synthesis not only validates BrightSteps' relevance but also positions it as an innovative contribution to the evolving DeFi landscape, addressing gaps in adoption and trust while advancing financial empowerment through blockchain technology.

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CHAPTER III

SOFTWARE METHODOLOGY AND SYSTEM ARCHITECTURE

This chapter outlines the software methodology and system architecture used to develop BrightSteps, a blockchain-integrated loan management application. It describes the systematic approach employed to design, implement, and evaluate the system, ensuring a secure, efficient, and user-centric lending solution. The chosen methodology provides a structured framework to manage the project lifecycle, support iterative improvements, and align each phase with the study's goals of improving financial accessibility and transparency in the decentralized finance (DeFi) ecosystem.

Software Methodology



Figure 1.0. The Agile Methodology

The Agile methodology emphasizes iterative development, teamwork, and continuous improvement to achieve high-quality software efficiently. Unlike traditional linear models (e.g., Waterfall), Agile breaks the project into smaller, manageable cycles called sprints, typically lasting 3-6 weeks. This approach allows the

development team to respond swiftly to changes, incorporate stakeholder feedback, and refine BrightSteps progressively. The methodology's focus on delivering functional increments ensures that critical features, such as Ethereum-based loan processing, are tested and deployed early and often. Figure 1.0 illustrates the Agile process, with each phase detailed below in the context of BrightSteps' development.

1. Plan. In the planning phase, the development team collaborates with TRUE COOP (Teachers, Retirees, and Urban Employees Cooperative) and cryptocurrency experts—to define BrightSteps' goals, scope, and requirements. Key features, including Ethereum wallet integration via MetaMask, loan application workflows, and loan tracking, are identified and prioritized in a product backlog. For example, the team prioritize secure user authentication and smart contract deployment over secondary features like advanced analytics. Each sprint begins with a planning session to select tasks from the backlog, ensuring alignment with the project's core objective: simplifying cryptocurrency lending for businesses.
2. Design. The design phase focuses on creating the system architecture and user interface (UI) for BrightSteps. The team designs a mobile-first UI using Jetpack Compose Kotlin to ensure consistency

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for Android, tailored to non-technical business owners. Wireframes outline intuitive screens for loan requests, eligibility checks, and repayment schedules. On the backend, vanilla PHP is architected to handle API calls to Ethereum nodes and manage off-chain data, such as user profiles. Security is paramount, with designs incorporating MetaMask integration for decentralized

transaction signing and Ethereum smart contracts for loan logic, ensuring a robust and transparent system.

3. Develop. During development, the team codes BrightSteps' features based on the design specifications. The mobile app, built with Kotlin, implements front-end functionalities like loan submission forms and ETH balance displays. The PHP backend handles server-side operations, such as querying Ethereum blockchain data via APIs (Infura) and processing loan statuses. Smart contracts, written in Solidity, are developed and deployed to the Ethereum network to automate loan disbursement and repayment. For instance, a smart contract might release ETH to a borrower's wallet once eligibility conditions are met, with repayment terms enforced on-chain.
4. Test. Testing ensures BrightSteps' functionality, security, and reliability. Unit tests verify individual components (e.g., MetaMask connectivity), while integration tests check end-to-end workflows, such as submitting a loan request and receiving ETH. Postman is used to validate API endpoints, ensuring seamless communication between the backend and Ethereum blockchain. Security audits focus on smart contract vulnerabilities (e.g., reentrancy attacks) and data

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encryption for user privacy. User Acceptance Testing (UAT) is conducted with a small group of business owners to evaluate the system's usability and functionality. Their feedback helps refine key features, such as the loan calculator and transaction confirmation process, ensuring a smoother user experience before full deployment.

5. Deploy. In the deployment phase, BrightSteps is released to be tested by

users. The mobile is not published considering it is a school project but smart contracts are deployed to the testnet (Ganache) for initial rollout.

6. Review. During the review phase, BrightSteps is evaluated with stakeholders after each sprint. User and client feedback helps identify key strengths, such as fast verification and loan approvals, and an intuitive user interface, and areas for improvement, like reducing transaction or gas fees. Performance is measured using metrics such as loan processing time and user satisfaction to ensure alignment with project objectives. Moreover, a System Usability Scale (SUS), a reliable tool for measuring usability, is also used as a key metric during the review phase. This helps assess user satisfaction and identify areas for improvement, ensuring that BrightSteps delivers an intuitive and efficient experience. This feedback is then used to refine the system, feeding into the planning phase to guide the next sprint.

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Software Architecture

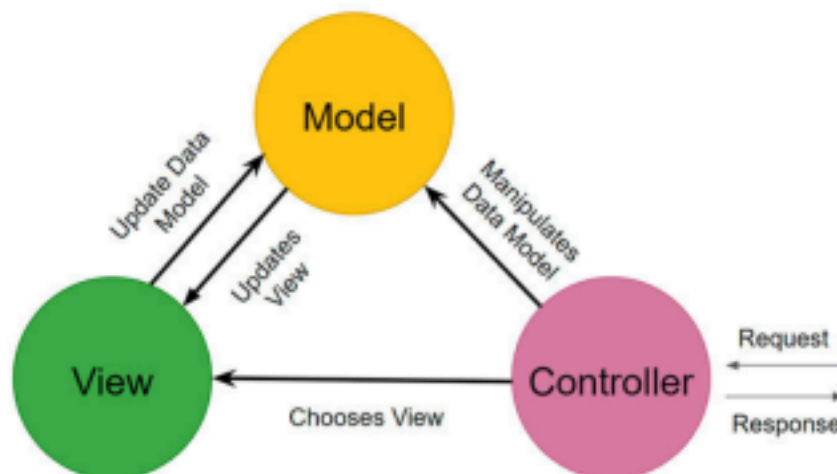


Figure 2.0. MVC Pattern

MVC stands for Model-View-Controller, and it's a software design pattern widely used for developing applications with user interfaces. It helps organize and structure the code to make it more manageable, maintainable, and scalable. Here's a breakdown of the three components:

1. Model: This represents the data and business logic of the application. The Model is responsible for fetching, updating, and processing data, usually by interacting with databases or APIs. It doesn't directly communicate with the user interface.
2. View: The View displays the data to the user. It's responsible for rendering the interface and ensuring the user can interact with it. However, the View doesn't handle any logic regarding how the data is manipulated.

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3. Controller: The Controller acts as the middleman between the Model and the View. It receives user input from the View, processes it (often involving the Model), and updates the View accordingly. Essentially, it handles the flow of data and logic in the application.

System Integration Method

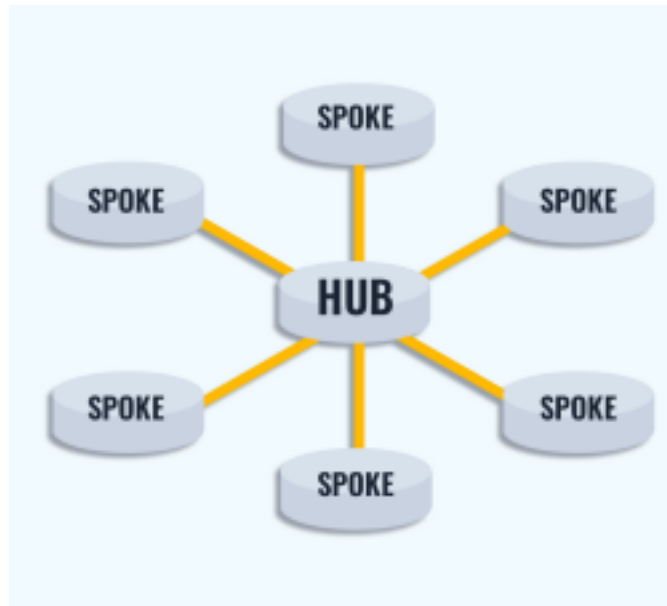


Figure 3.0. Hub and Spoke System Architecture

Complementing the Agile Methodology, the development of BrightSteps employs the Hub and Spoke system integration method, an advanced architectural approach designed to address the limitations of point-to-point integration and prevent the inefficiencies of a star or "spaghetti" configuration. In this model, a central hub, functioning as a message broker, manages all interactions between subsystems, eliminating direct communication among components such as the mobile application, web admin interface, and blockchain infrastructure. This architecture was

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selected for its balance of simplicity, scalability, and compatibility with BrightSteps' Ethereum-based lending requirements.

Population and Sample

To assess BrightSteps' performance, scalability, and user satisfaction, this study defines a target population and calculates a sample size for testing and

validation. This section outlines the population, and to determine a statistically appropriate sample, and links the sample size to the system's scalability requirements.

The target population for BrightSteps consists of a focused group of users within the lending ecosystem, specifically members of TRUE COOP (Teachers, Retirees, and Urban Employees Cooperative) involved in the project's testing phase. For this study, the population is limited to a manageable size of 52 individuals, representing cooperative members seeking loans and administrators managing loan processes. This smaller population reflects a controlled test environment suitable for a school project, focusing on active TRUE COOP participants willing to engage with the blockchain-integrated application via MetaMask. The total population is thus defined as $N = 52$.

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Sampling Technique

A simple random sampling technique is employed to select the 52 participants from the population of 60. Given the small population size, this method ensures each member has an equal chance of selection, capturing diverse user experiences without requiring stratification. The sample includes both borrowers and administrators to reflect the cooperative's operational roles, offering comprehensive insights into the system's performance.

$$n = \frac{N \cdot z^2 \cdot p \cdot q}{1 + z^2 \cdot p \cdot q}$$

Where:

- n = sample size
- N = total population size (50)
- e = margin of error (set at 0.05 for a 95% confidence level)

Substituting the values:

$$\diamond \diamond = \frac{60}{1 + 60(0.05^2)}$$

$$\diamond \diamond = \frac{60}{1 + 0.15}$$

$$\diamond \diamond = \frac{60}{1.15}$$

$$\diamond \diamond = 52.17$$

Thus, the calculated sample size is approximately 52 users. This sample represents the number of users or transactions required to assess BrightSteps' functionality, security, and scalability during development.

Instrumentation

This study employed a questionnaire as the primary data collection tool to evaluate the usability, functionality, and user satisfaction of BrightSteps. After reviewing instruments from related studies in financial technology and usability testing, the developers adopted a self-administered questionnaire incorporating the standardized System Usability Scale (SUS). This approach ensured the collection of data necessary to address the study's objectives of assessing BrightSteps' effectiveness in providing a secure, transparent, and efficient lending solution within the decentralized finance (DeFi) ecosystem.

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The questionnaire developed by the proponents consisted of two parts. The first part focused on gathering essential background information about the respondents, including demographic and contextual profiles, such as the respondent's name, role within TRUE COOP (e.g., borrower or administrator), and prior experience with digital financial tools or cryptocurrency (e.g., familiarity with Ethereum or MetaMask). This section helped establish the respondents' characteristics to contextualize their usability feedback on BrightSteps.

The second part utilized the System Usability Scale (SUS) to assess the respondents' experience with BrightSteps' usability. The SUS is a standardized instrument comprising 10 specific statements, each rated on a 5-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). These statements evaluate the system's ease of use, learnability, and overall user satisfaction. The SUS questions, adapted to BrightSteps, include:

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to use this system.
5. I found the various functions in this system were well integrated. 6. I thought there was too much inconsistency in this system.

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7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very inconvenient to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

Each respondent rated these statements based on their interaction with BrightSteps, with scores ranging from 1 (lowest) to 5 (highest). The SUS scoring process involves converting responses into a final score out of 100, where odd-numbered items (1, 3, 5, 7, 9) contribute positively and even-numbered items (2, 4, 6, 8, 10) are reversed. Respondents completed the questionnaire in approximately 10 to 15 minutes after using the application during the Test and Deploy phases of the Agile methodology. This standardized format allowed the developers to benchmark BrightSteps' usability against industry norms and quantify user perceptions effectively.

To establish the content validity and accuracy of the research instrument, the

developers relied on the SUS's established reliability as a validated tool, widely used in usability studies. Additionally, they sought feedback from experts to ensure the questionnaire's applicability to BrightSteps' context. This included a cryptocurrency expert familiar with Ethereum and smart contracts, a software development mentor overseeing the Agile process, and a TRUE COOP representative to confirm relevance to

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cooperative lending needs. Their input refined the administration process (e.g., instructions for respondents) and ensured the SUS questions were appropriately contextualized for a blockchain-based lending application, maintaining alignment with the study's goals of enhancing financial accessibility and trust.

Data Gathering Procedure

The Republic Act No. 10173, also known as the Data Privacy Act of 2012, is implemented to safeguard individuals' personal information while facilitating the free flow of communication. This act underscores the critical role of information and communications technology and mandates the protection of personal data privacy. In compliance with this legislation, the developers ensured that all data collection for BrightSteps—a blockchain-integrated loan management application—was conducted securely, guaranteeing the confidentiality and protection of participants' personal information throughout the process. The developers initiated the data gathering by drafting a formal letter requesting approval to conduct the study. Upon receiving approval, the developers prepared a consent form detailing the purpose, procedures, and data privacy measures of the study. This form was presented to the participants

to obtain their informed consent before proceeding with data collection, ensuring transparency and voluntary participation. The structured questionnaire, incorporating the System Usability Scale (SUS) as outlined in the Instrumentation section, was

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distributed in printed form to the sample of 52 TRUE COOP members selected via simple random sampling from a population of 60. These participants included borrowers and administrators who interacted with BrightSteps during the Test and Deploy phases of the Agile methodology. The data collection occurred after participants used the BrightSteps application on the Ganache testnet, allowing them to experience its features (e.g., loan requests, Ethereum transactions via MetaMask). The questionnaires were administered in a controlled setting, such as a cooperative meeting or a designated testing session, and collected immediately upon completion to maintain data integrity. Each respondent took approximately 10 to 15 minutes to answer the SUS-based questionnaire, rating their experience with BrightSteps' usability. Upon gathering all responses, the developers meticulously assessed and analyzed the data using statistical methods. Specifically, SUS scores were calculated by converting individual responses into a composite score out of 100 (as per SUS methodology: odd items adjusted by subtracting 1, even items reversed by subtracting from 5, summed, and multiplied by 2.5). This analysis evaluated BrightSteps' effectiveness in delivering a secure, transparent, and user-friendly lending solution.

Statistical Treatment of System Usability Scaling's Data

In this study, the System Usability Scale (SUS) was employed to assess the

usability of the system in question. The SUS, a standardized tool

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for evaluating the effectiveness, efficiency, and satisfaction of system design, produces scores ranging from 0 to 100, where higher scores indicate greater usability.

1. Data Collection and Scoring

The SUS data were collected through user surveys, consisting of 10 Likert-scale questions ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). Each participant's responses were transformed into individual SUS scores using the established scoring formula. Specifically:

◆◆ = Total Sum of the Odd Numbers - 5

◆◆ = Total Sum of the Even Numbers Subtracted from 25

The resulting values were summed and multiplied by 2.5 to normalize the scores to a 0–100 scale.

◆◆ + ◆◆ = ◆◆ multiplied by 2.5 = Individual SUS Score

The overall System Usability Scale (SUS) score was calculated by obtaining the arithmetic mean of all individual SUS scores. This approach ensures a comprehensive representation of user feedback, encapsulating the collective usability assessment across all participants.

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$$\begin{array}{c} \text{◆◆} \\ \text{◆◆} \end{array} = 1 \begin{array}{c} \text{◆◆} \\ \text{◆◆} \end{array} = 1$$

$$\sum \diamond \diamond \diamond \diamond$$

Where:

$\diamond \diamond$ = arithmetic mean

$\diamond \diamond$ = number of values

$\diamond \diamond$ = data set values
 $\diamond \diamond$

2. Data Interpretation

The results of statistical analyses were interpreted in the context of usability benchmarks and prior literature. A SUS score above 68 was considered good, while scores above 80.3 signified excellent usability.

Score	Grade	Interpretation
>80.3	A	Excellent
68 - 80.3	B	Good
68	C	Okay
51 - 68	D	Poor
< 51	E	Awful

3. Result

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BrightSteps collected responses from a total of 53 participants, resulting in an arithmetic mean usability score of 87.38. This score falls within the "Excellent" range, reflecting high levels of user satisfaction with the system's functionality, design, and ease of use. The data suggests that BrightSteps effectively meets

user expectations and delivers a well-optimized experience, underscoring its strengths in usability and engagement. This exceptional score emphasizes the platform's potential to set a standard for future systems in its domain.

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CHAPTER IV

INTERFACE

This chapter provides an overview and analysis of the application and web interface, focusing on its design, functionality, and user interaction. It highlights the main components such as navigation menus, dashboards, and input forms, while emphasizing their role in creating a seamless workflow for users. The chapter also reviews usability testing results, discussing metrics like efficiency and satisfaction, and concludes with recommendations for improving accessibility and responsiveness to enhance overall user experience.

Web Features and Pages



Figure 4.0. Admin Login Page

The "Admin Login" page serves as the starting interface for administrators, featuring the BrightSteps logo as a welcoming element. It includes a login form with labeled text fields for "Email" and "Password," and a "Remember Me" checkbox that allows automatic login whenever the site is accessed. A default admin account is pre-created in the database, enabling quick access by entering pre-set credentials.

The page also offers a "Forgot Password" option to reset the password if needed. Below the form, there's a "Login" button to confirm the credentials and navigate directly to the dashboard. The overall design is responsive and ensures ease of use for administrators.

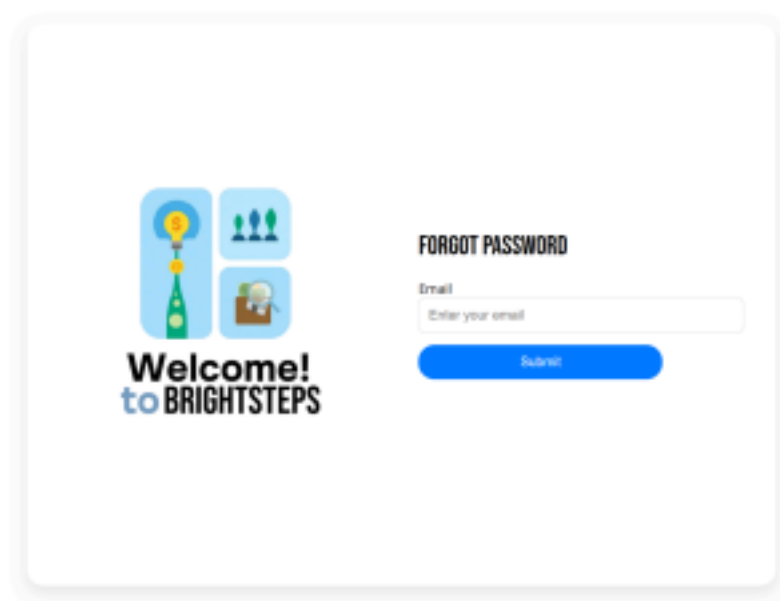


Figure 4.1. Forgot Password Page

When the 'Forgot Password' button on the 'Login' page is clicked, it redirects to the 'Forgot Password' page. Here, the admin begins the

password reset process by entering their 'Email' for verification. A 'Submit' button is available, and once clicked, it sends an OTP (One-Time Password) to the admin's email for authentication.

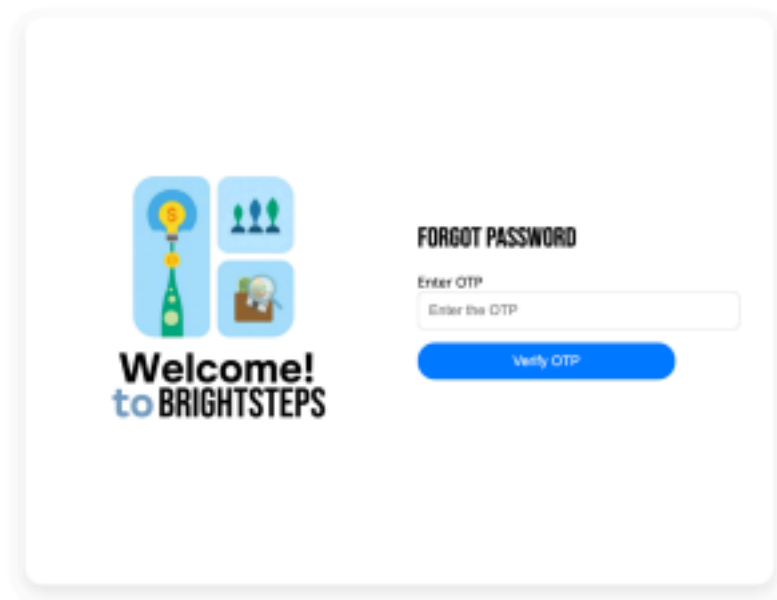


Figure 4.2. Verify OTP Page

Once an OTP (One-Time Password) is sent to the admin's email, it must be entered on the Verify OTP Page to confirm their identity before proceeding with the password reset process.

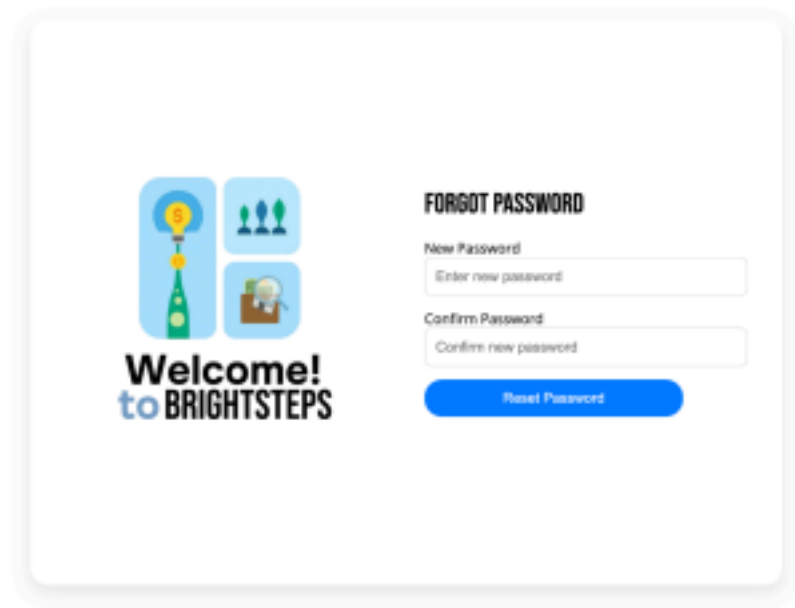


Figure 4.3. Change Password Page

The Change Password screen allows the admin to set a new password after successfully verifying the OTP. On this screen, the admin must enter and confirm their new password to ensure accuracy. Once submitted, the new password will replace the old one, granting the admin access to log in with their updated credentials.

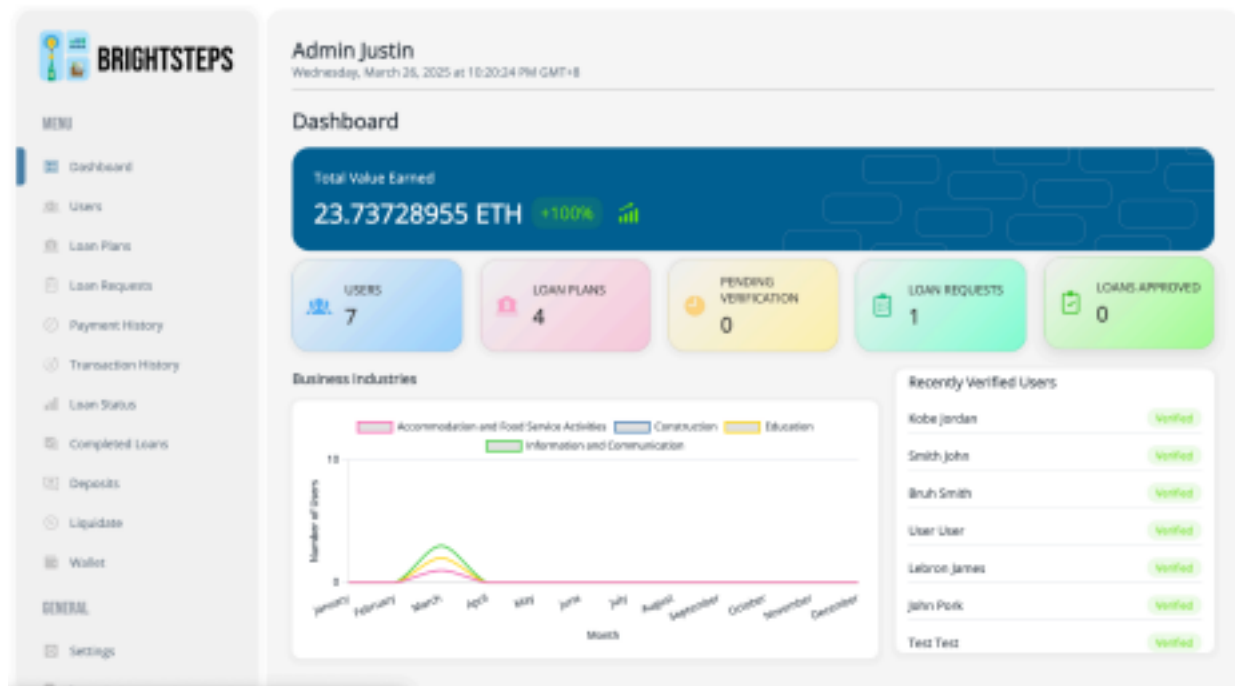


Figure 5.0. Dashboard Page

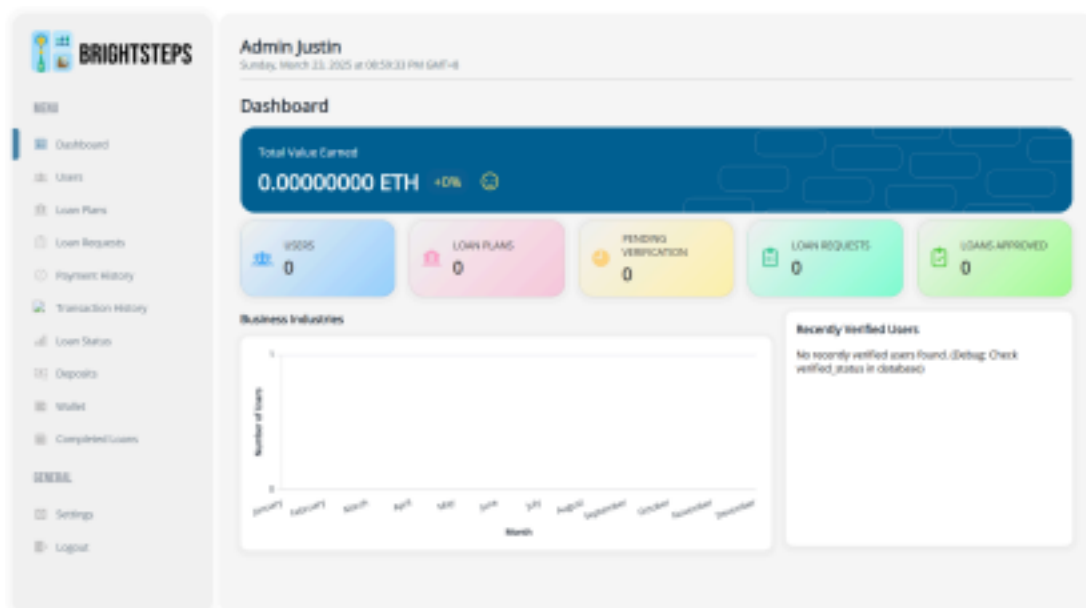
The 'Dashboard' page offers an overview of BrightSteps' data. On the left side, there are navigation links divided into two categories: 'Menu,' which leads to different admin web pages, and 'General,' which provides access to the 'Settings' page and the logout option.

At the top of the main Dashboard section, the admin's name and the real-time date and time are displayed. Below this, a section highlights the total value earned, accompanied by buttons that link to other pages:

- The 'User' button displays the total number of users and redirects to the 'User' page.
- The 'Loan Plans' button shows the number of loan plans and links to the 'Loan Plans' page.

- The 'Pending Verification' button lists users awaiting verification and navigates to the 'Users' page with a 'Pending' filter applied
- The 'Loan Request' button reflects the number of loan requests and redirects to the 'Loan Requests' page.
- The 'Loan Approved' button shows approved loan users and leads to the 'Loan Status' page.

Below these buttons, a line graph illustrates the distribution of verified users across business industries. Adjacent to the graph, the 'Recently Verified Users' tab presents a list of users who were recently verified



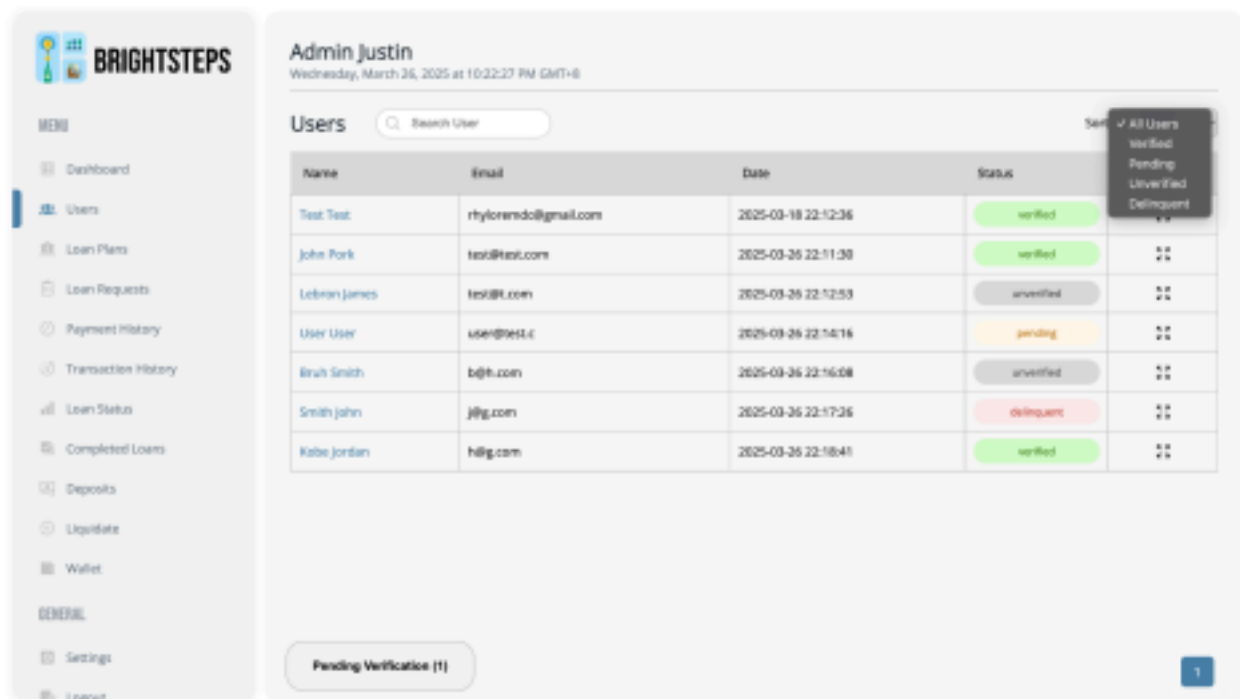


Figure 6.0. Users Page

The 'Users' page displays a complete list of BrightSteps users. On the left side, there are navigation links divided into two categories: 'Menu,'

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which leads to different admin web pages, and 'General,' which provides access to the 'Settings' page and the logout option.

At the top of the main Users section, the admin's name and the real-time date and time are displayed. there's a search bar to help the admin find specific users quickly. On the right side, a dropdown menu is available, allowing the admin to sort the user list as needed.

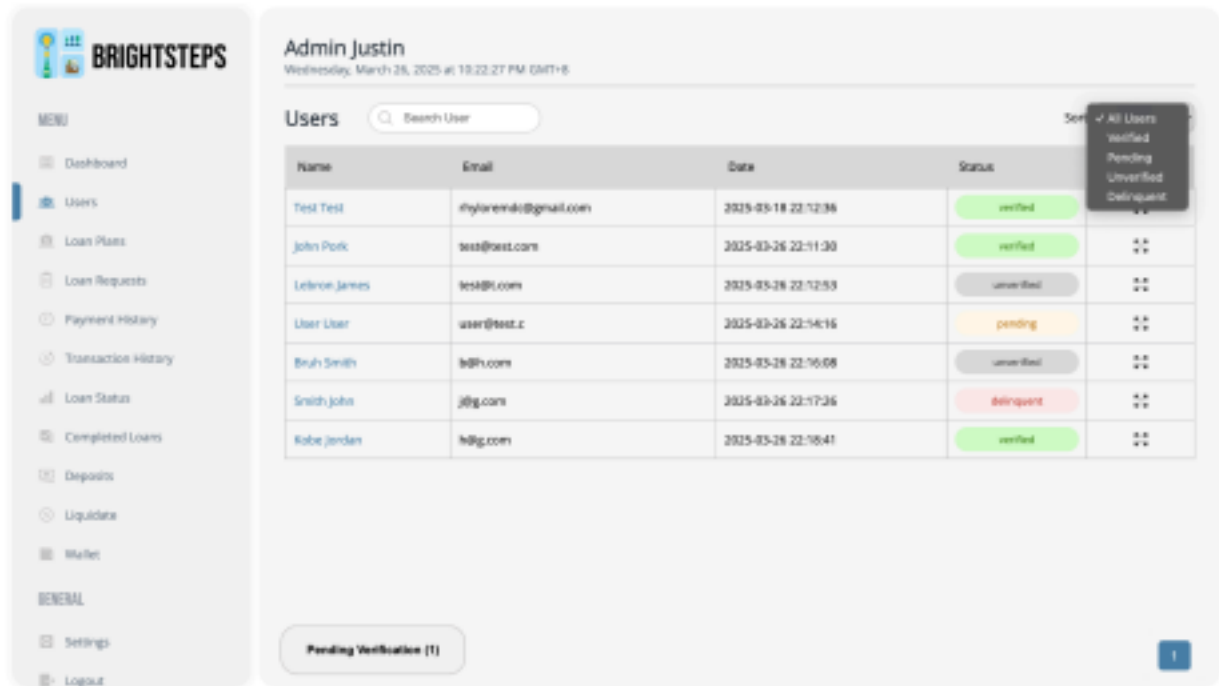


Figure 6.1. Users Page with Dropdown list

The 'Users Page with Dropdown List' allows the admin to sort users based on specific categories, including 'All Users,' 'Verified,' 'Pending,' 'Unverified,' and 'Delinquent' users. This functionality helps the admin manage and navigate user data efficiently.

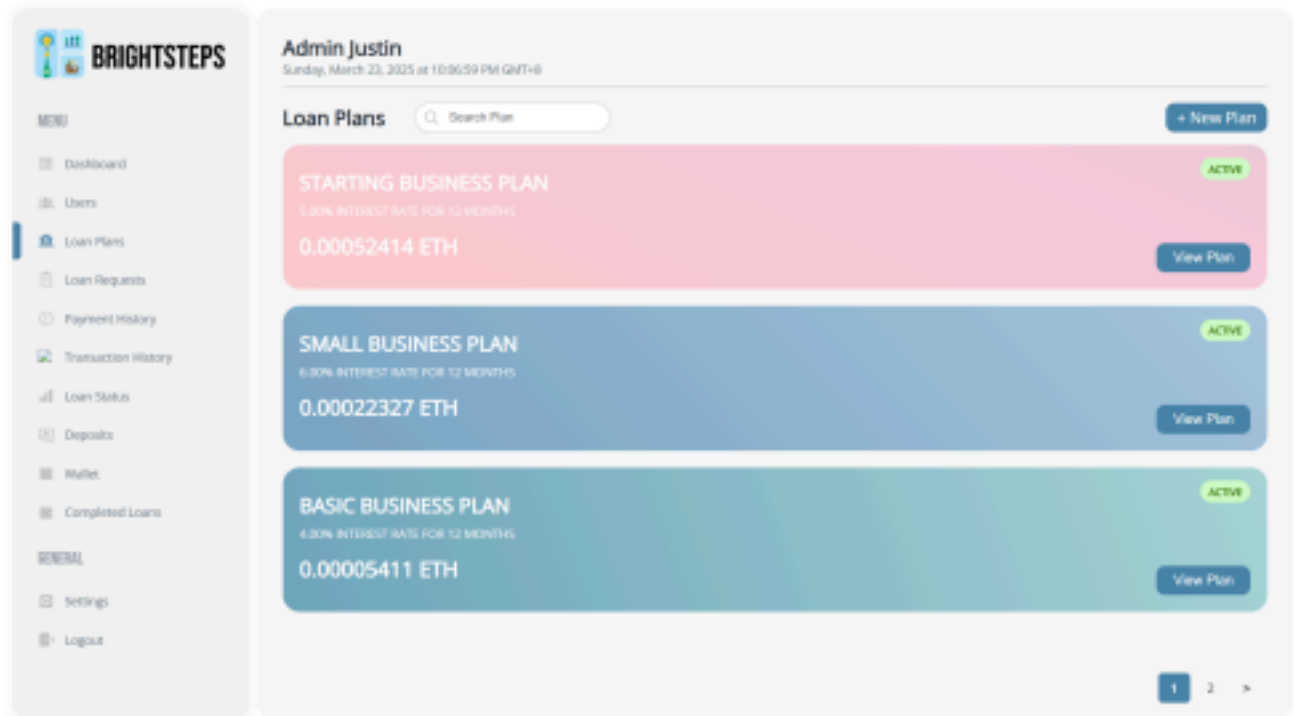


Figure 7.0. Loan Plans Page

The 'Loan Plans' page provides an overview of all created loan plans. On the left, there are navigation links organized into two sections: 'Menu' for accessing various admin web pages and 'General' for navigating to the 'Settings' page or logging out.

At the top of the Loan Plans section, the admin's name and the current date and time are displayed. A search bar is available to help the admin quickly find specific loan plans. On the right, a 'New Plan' button opens a modal for creating a new loan plan when clicked. Below this section, the created loan plans are displayed, with three plans shown per page. Each loan plan lists its title, interest rate, duration, loan amount, status, and

includes a 'View Plan' button that opens a modal with more details, allowing the

admin to update the plan if needed. Pagination is available at the bottom-right corner of the page to navigate through additional loan plans if there are more than three.

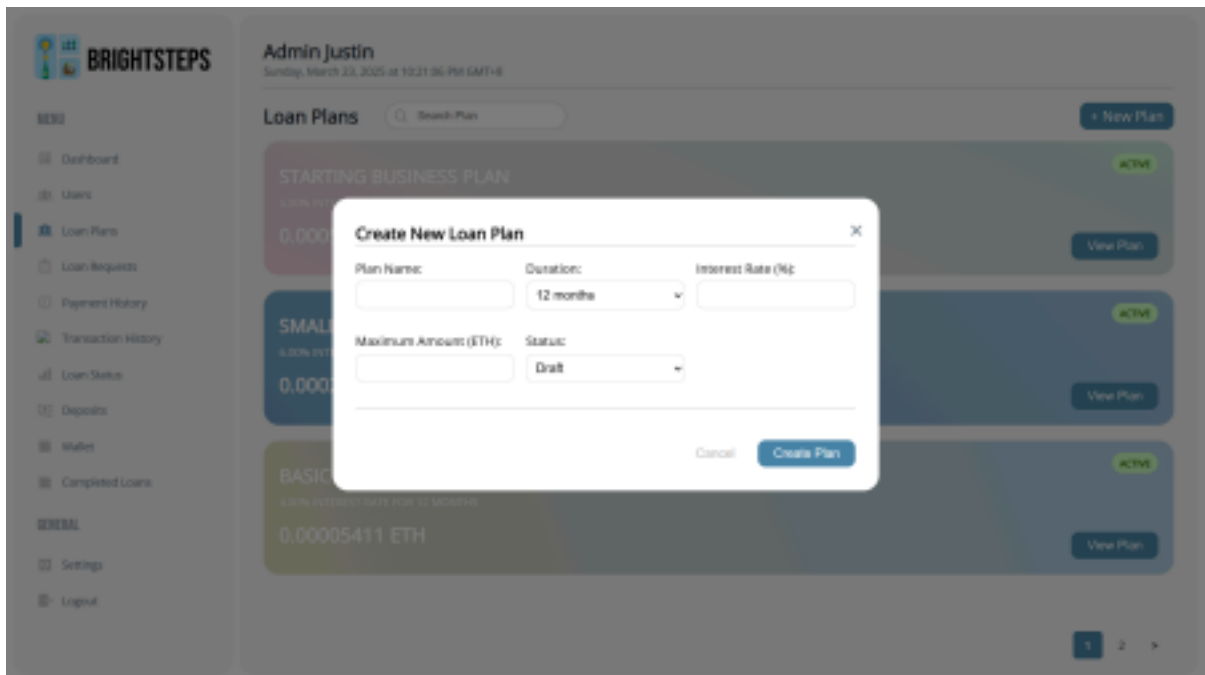


Figure 7.1. Loan Plans Page with New Plan Button Clicked

When the 'New Plan' button is clicked on the 'Users' page, a modal appears with fields to input details such as Plan Name, Maximum Amount (ETH), Duration, Status, and Interest Rate (%). Below these fields, there are buttons to either cancel the creation process or finalize and create the plan.

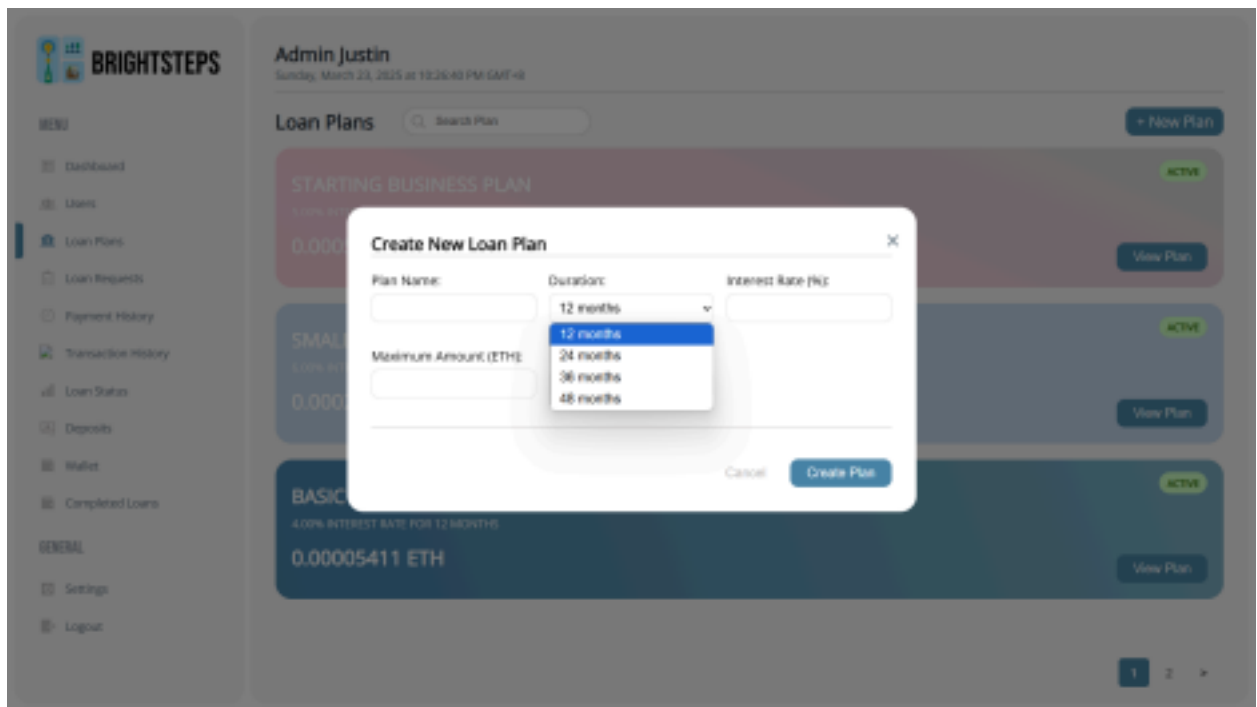


Figure 7.2. Create New Loan Plans with Duration Dropdown

This displays the list of available loan duration options in months, which the admin can select while setting up a new loan plan.

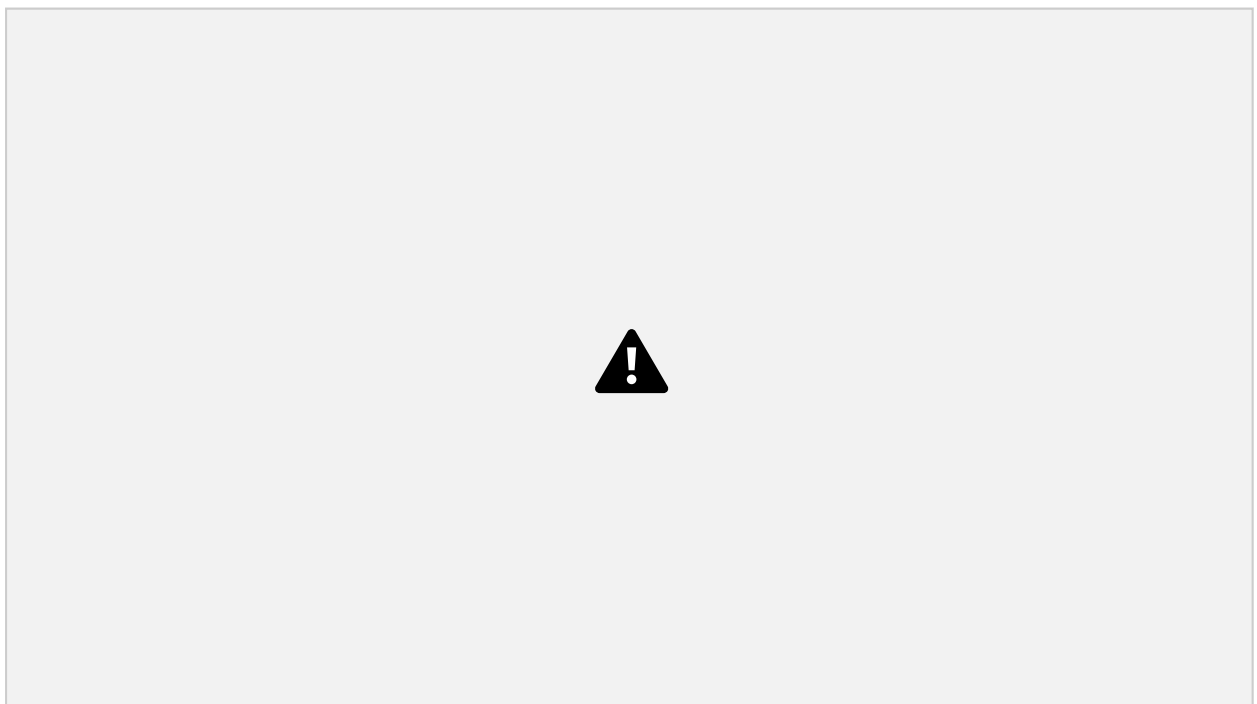


Figure 7.3. Create New Loan Plans with Status Dropdown

This shows the status of loan plans after creation. Loan plans marked as 'Draft' are visible only to the admin, who can update their status to either 'Active' or 'Inactive.' Loan plans with an 'Active' status are accessible to users, allowing them to view and apply for these plans. However, plans with an 'Inactive' status are visible only to users who already have ongoing loans under those plans, while other users cannot view or interact with them.

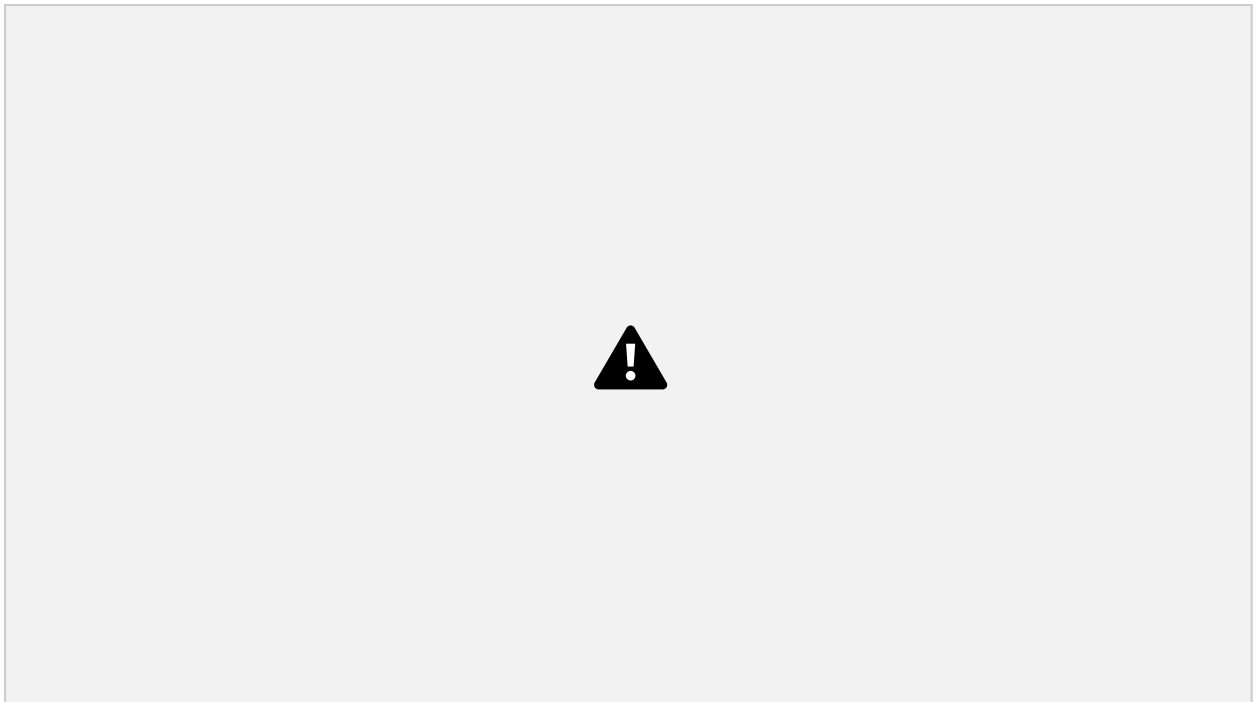


Figure 7.4. Loan Plans Page with View Plan Button Clicked

This modal presents detailed information about a loan plan and provides options for the admin to update it. At the bottom, there are two buttons: 'Delete Plan,' which removes the loan plan, and 'Save Changes,' which applies any updates made

by the admin.

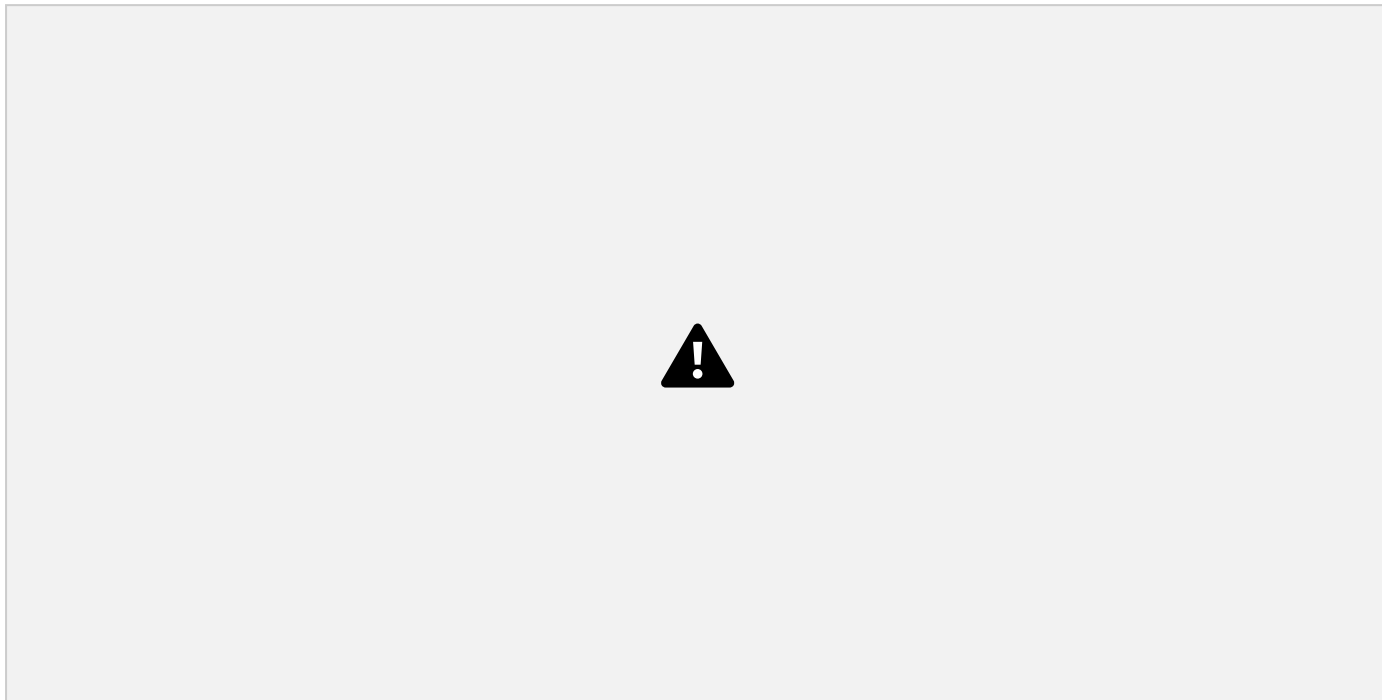


Figure 8.0. Loan Requests Page

The 'Loan Requests' page provides an overview of all the users who applied for a loan. On the left, there are navigation links organized into two

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sections: 'Menu' for accessing various admin web pages and 'General' for navigating to the 'Settings' page or logging out.

At the top of the Loan Requests section, the admin's name and the current

date and time are displayed. A search bar is available to help the admin quickly find specific users.

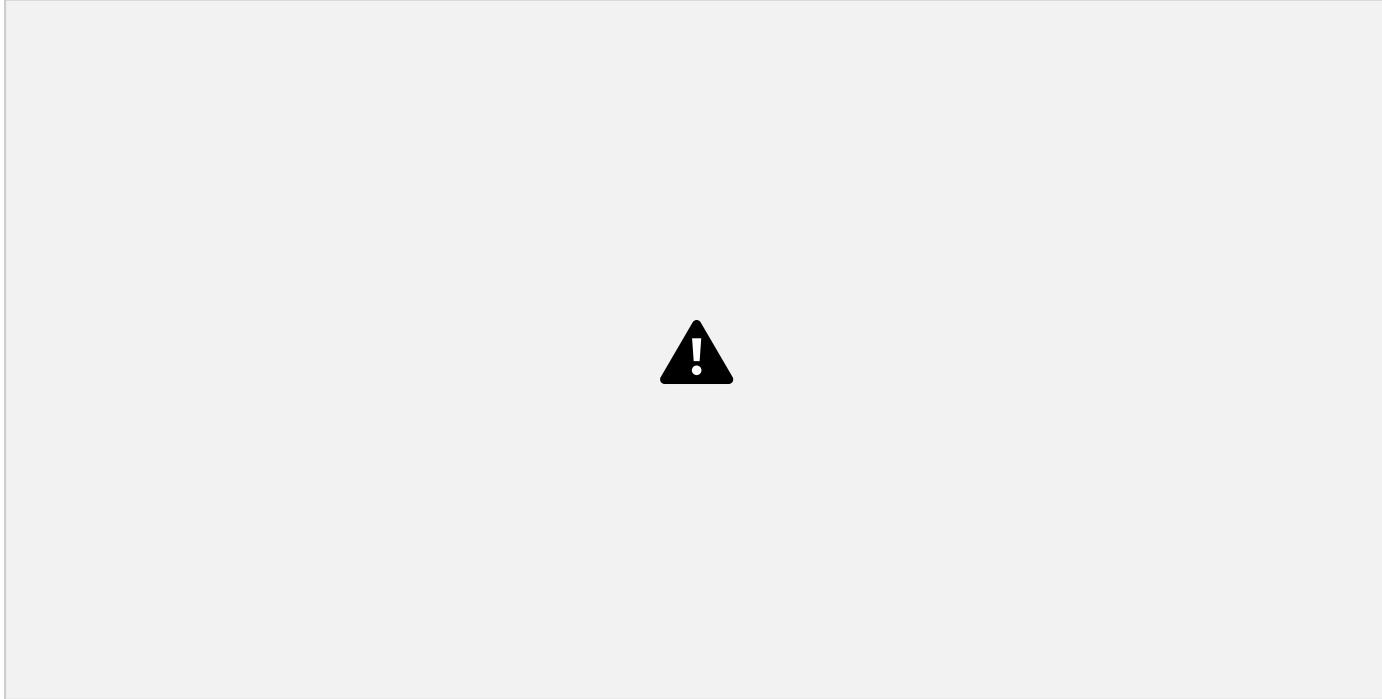


Figure 9.0. Payment History Page

The 'Payment History' page offers a summary of all users who make monthly loan payments. On the left, navigation links are divided into two categories: 'Menu,' which leads to various admin web pages, and 'General,' which provides access to the 'Settings' page and the logout option.

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At the top of the Payment History section, the admin's name and the current date and time are shown. A search bar is provided to help the admin locate specific users efficiently, and next to it is a filter dropdown menu for sorting users

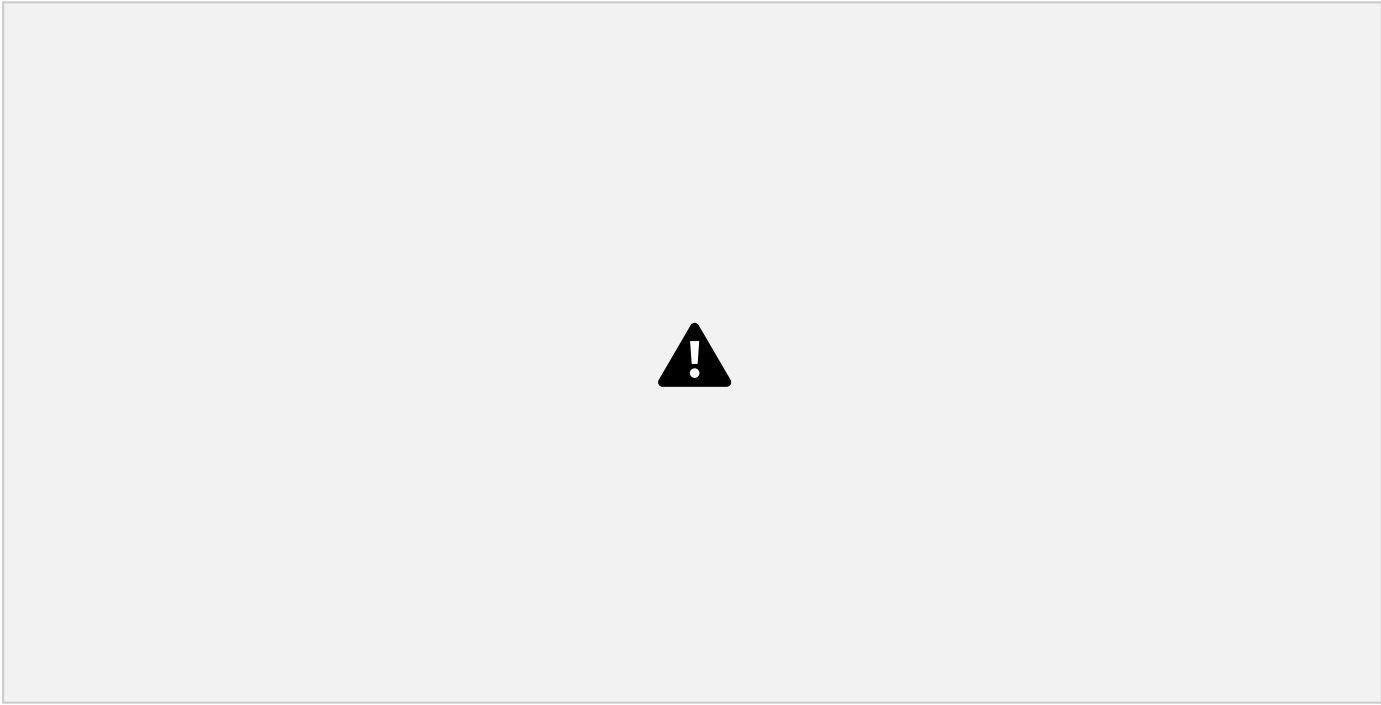


Figure 9.1. Payment History Page with Filter Dropdown List

Clicking this option will sort users based on the selected criteria, either by 'Repayment' or 'Deposits Sent.'

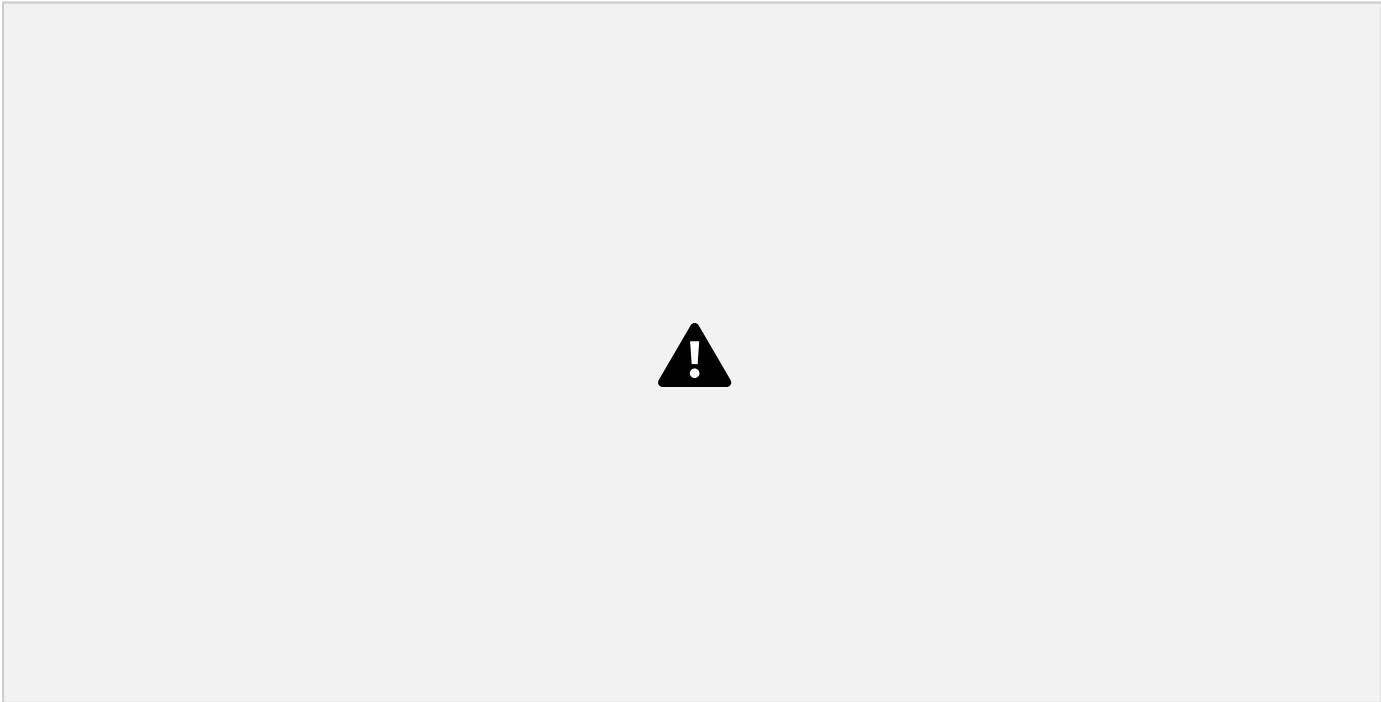


Figure 10.0. Transaction History Page

The 'Transaction History' page offers a summary of all the transactions made. On the left, navigation links are divided into two categories: 'Menu,' which leads to various admin web pages, and 'General,' which provides access to the 'Settings' page and the logout option.

At the top of the main Transaction History section, the admin's name and the current date and time are shown. A search bar is provided to help the admin locate specific users efficiently, and next to it is a filter dropdown menu for sorting users

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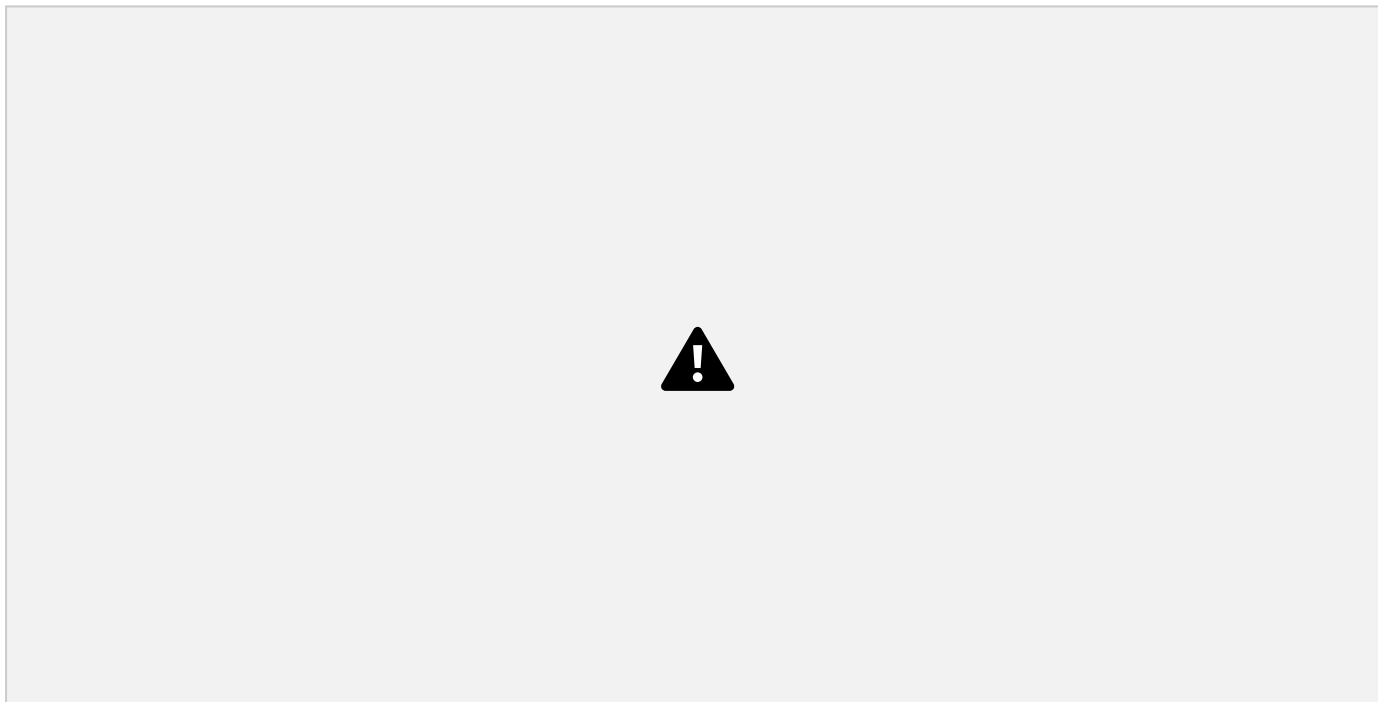


Figure 11.0. Loan Status Page

The 'Loan Status' page offers a summary of all users who applied for a loan. On the left, navigation links are divided into two categories: 'Menu,' which leads to various admin web pages, and 'General,' which provides access to the 'Settings' page and the logout option.

At the top of the main Loan Status section, the admin's name and the current date and time are shown. A search bar is provided to help the admin locate specific users efficiently, and next to it is a filter dropdown menu for sorting users.

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Figure 11.1. Loan Status Page with Filter Dropdown List

Clicking this option will sort users based on the status selected. Either by 'All Status', 'Active', which is the approved loan requests, 'Rejected', and 'Completed'.

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Figure 12.0. Deposits Page

The 'Deposits' page offers a summary of all users who are available to have deposits. On the left, navigation links are divided into two categories: 'Menu,' which leads to various admin web pages, and 'General,' which provides access to the 'Settings' page and the logout option.

At the top of the main Deposits section, the admin's name and the current date and time are shown. A search bar is provided to help the admin locate specific users efficiently.

Figure 13.0. Wallet Page Part 1

The 'Wallet' page offers a summary of all finances. On the left, navigation links are divided into two categories: 'Menu,' which leads to various admin web pages, and 'General,' which provides access to the 'Settings' page and the logout option.

At the top of the main Wallet section, the admin's name and the current date and time are shown. On the top-right side of the screen is a button labeled 'Connect Metamask' which is needed to fetch the total amount of the admin's wallet. Below those is the section where the total amount and value of the admin's wallet will be displayed. Below is an overview of the wallet statistics like the 'Yearly Earnings and Monthly Earnings'.

Figure 13.1. Wallet Page Part 2

Beneath the 'Yearly Earnings and Monthly Earnings' section lies a line graph illustrating the growth rate of average earnings. Below this is another line graph that displays the trend of ETH prices. Positioned to the right is a pie chart that represents loan repayment statistics, highlighting the proportions of 'on-time' and 'late' loans.



Figure 14.0. Completed Loans Page

The 'Completed Loans' page offers a summary of all users who completed their loan. On the left, navigation links are divided into two categories: 'Menu,' which leads to various admin web pages, and 'General,' which provides access to the 'Settings' page and the logout option.

At the top of the main Completed Loans section, the admin's name and the current date and time are shown. A search bar is provided to help the admin locate specific users efficiently.

This page also allows the admin to refund the user's deposit once they have completed their loan.



Figure 15.0. Settings Page

The 'Settings' page provides an overview of the available settings for BrightSteps. On the left, there are navigation links organized into two sections: 'Menu' for accessing various admin web pages and 'General' for navigating to the 'Settings' page or logging out.

At the top of the Settings section, the admin's name and the current date and time are displayed. A maintenance mode feature is provided, allowing the admin to enable or disable it as needed. When activated, it temporarily restricts access to the app on the mobile/user side until the admin deactivates the maintenance mode.

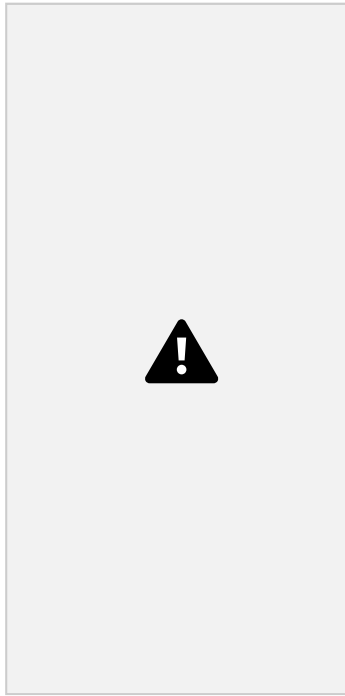


Figure 16.0. Splash Screen

Upon opening the app, the splash screen displays for a few seconds, showcasing the app's logo and branding. It then transitions to the landing page unless the app is in maintenance mode, in which case a maintenance notice is shown instead.

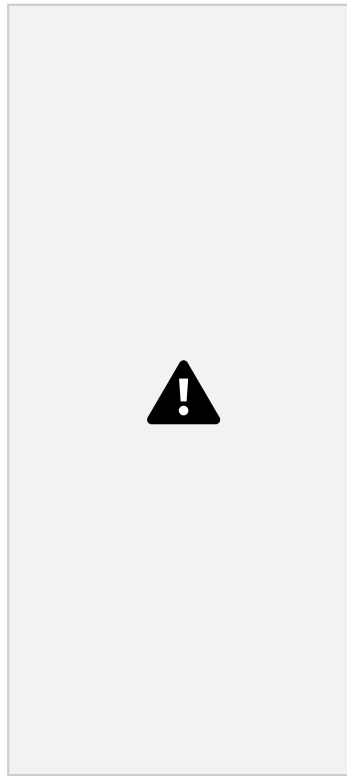
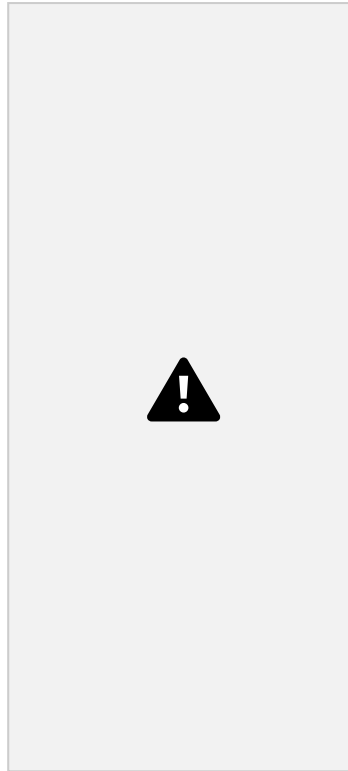


Figure 17.0. Landing Screen

The landing screen serves as the app's entry point, featuring a clean and user-friendly interface. It presents two primary buttons: "Log In" for returning users to access their accounts and "Create an Account" for new users to sign up. This page ensures seamless navigation to the app's main functionalities.

Figure 17.1. Login Screen
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The login screen allows returning users to securely access their accounts by entering their credentials. It includes fields for email and password, along with options for password recovery and a text button that navigates the user to the

registration screen in case they made the mistake of pressing the log in button on the landing screen instead of the 'Create an Account' button.

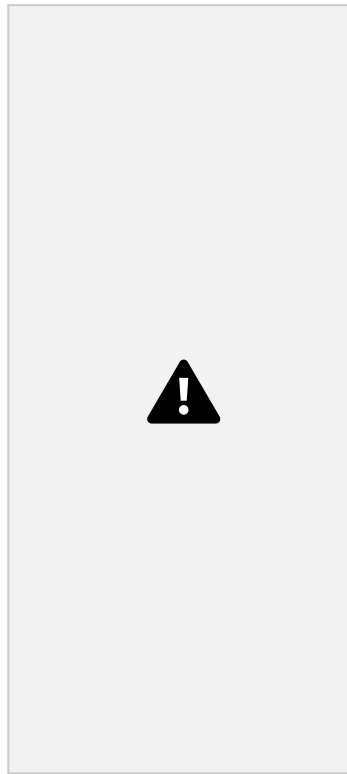
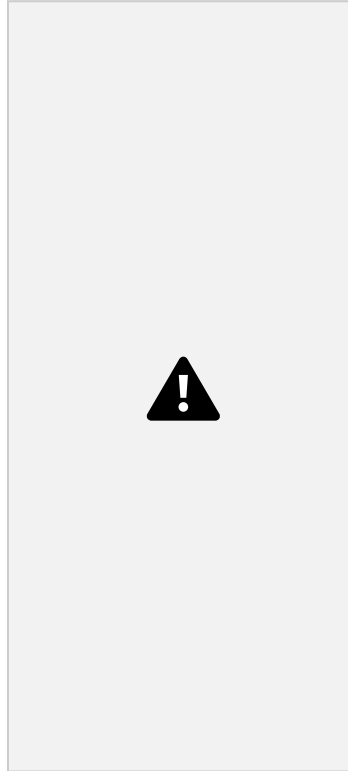


Figure 17.2. Registration
Screen
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The register screen enables new users to create an account by providing

necessary details such as name, email, and password, which is necessary for the profile setup, ensuring a smooth onboarding process.



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Figure 17.3. Forgot Password Screen

The forgot password screen helps users reset their password by entering their registered email. A One-Time Password (OTP) is sent to verify their identity, which is required to be validated on the OTP verification screen, guiding them through the password recovery process.

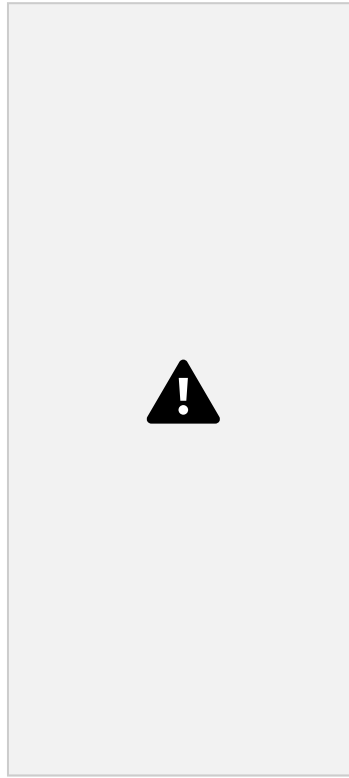
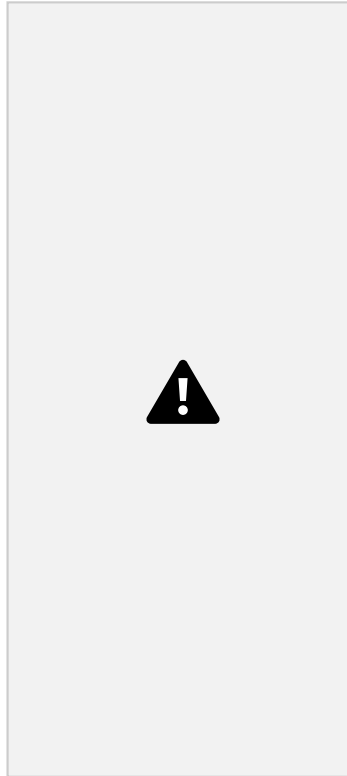


Figure 17.4. OTP Verification Screen

The OTP verification screen prompts users to enter a one-time password (OTP) sent to their registered email. This step ensures security and confirms the user's identity before proceeding with password reset.

Figure 17.5. Change Password
Screen
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The change password screen allows users to set a new password after successful OTP verification. It includes fields for entering and confirming the new

password.

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Figure 18.0. Home Screen

The Home Screen provides users with a centralized view of their loan details and available loan options. At the top, a hamburger menu icon allows access to the side navigation drawer for seamless app navigation. The dashboard prominently displays the user's current loan status, including the outstanding balance and monthly payment details. Below, a list of available loan plans is presented, enabling users to explore and apply for new loan options effortlessly.

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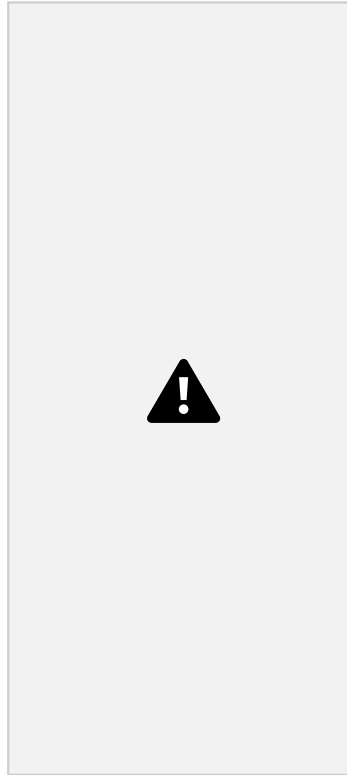


Figure 18.1. Home Screen with Side Navigation Menu Open

When the hamburger menu icon is tapped, the side navigation menu slides open, providing quick access to essential app features. The menu includes options for managing the user account, such as User Account and Change Password. Users can also navigate their loan details through Current Loan, Loan History, and Apply Loan. Payment-related options include Make Payment and Payment History for tracking transactions. Additionally, support and information sections like Contact Us, FAQs, and Rules and Policies are available. At the bottom, a Logout button allows users to securely exit the app.

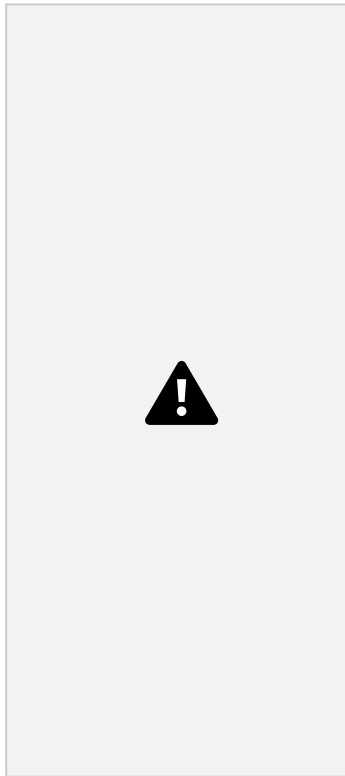


Figure 18.2. Home Screen with Active Loan

In this view, the Current Loan section displays the user's outstanding balance and the remaining months left for repayment. The Monthly Payment section shows the fixed amount the user needs to pay each month. When a loan plan is tapped, the user is redirected to the details of their current loan plan for further information.

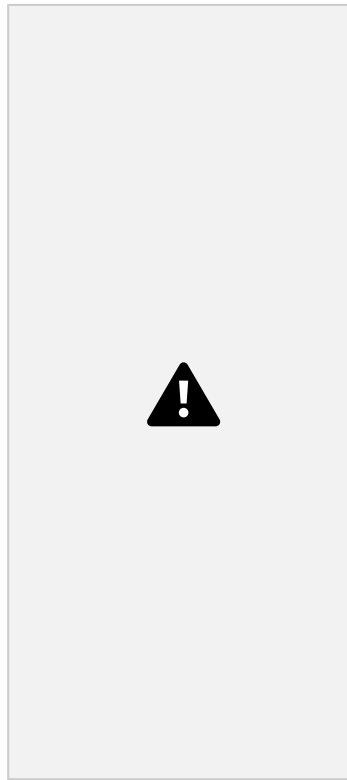


Figure 19.0. Loan Application Screen

The Loan Application Screen presents the selected Loan Plan's Informative Loan Card, detailing the loan amount, interest rate, and duration. Below, a text box allows users to specify their reason for loaning. Additionally, the screen includes terms and conditions that users must review and agree to before proceeding with their loan application, and the estimated monthly repayment for the loan.

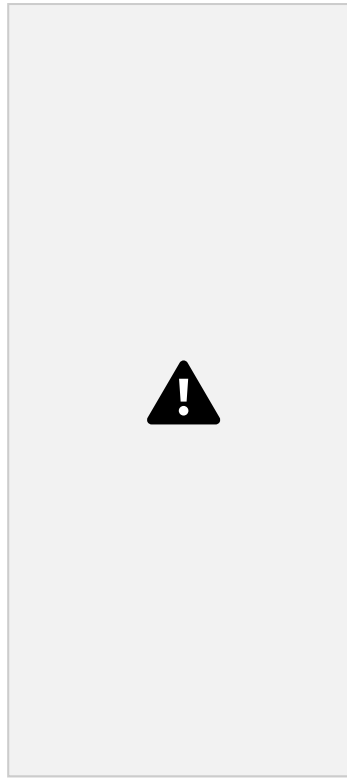


Figure 19.1. Deposit Alert Dialog

Upon opening the loan application screen, an alert dialog already notifies the user about the requirement of an initial deposit for the loan. This dialog contains information about how much is the required deposit for the loan, and how much can be borrowed from the deposit.